

Income Composition and National Economic Evaluations in the 2012 and 2020 Elections

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Abstract

Equal growth in aggregate income can reflect different economic circumstances. Does its composition distinguish national economic evaluations? This study links the 2012 and 2020 Cooperative Congressional Election Studies to state labor conditions and income accounts, expressing earnings, property income, and transfers as contributions to a common total. Separate weighted ordinal models compare their associations with favorable national judgments. Adjusted intervals do not establish the joint expectation of positive labor and earnings associations. Common-coefficient restrictions are not rejected, but most direct component comparisons remain unresolved. Two primary comparisons in 2020 meet a specified practical-similarity criterion; one loses that classification under state-deletion uncertainty. These findings distinguish uncertainty about income composition from evidence that components have similar local associations. They support a bounded measurement comparison, without establishing a general total-income account or identifying how individuals interpret economic conditions.

Keywords: economic evaluations; income composition; retrospective accountability; contextual information; survey inference

Introduction

An economy can offer more resources without offering more work. Growth in personal income can consist of earnings, dividends and rent, public benefits, or combinations of these sources. Equal increases in total income can therefore accompany different changes in employment and different distributions of resources among residents. For research on economic evaluations, the distinction raises a measurement question: how are pre-election state labor-market and income-component changes associated with national economic evaluations? A total-income measure summarizes resources, but the circumstances behind that total may differ. Its adequacy as a description of economic judgments depends on whether the sources of growth have similar associations with those judgments.

I compare earnings, property income and transfers as additive contributions to the same change in total income, each measured relative to prior total income. This common scale makes explicit the equal-association restriction imposed by replacing the components with their sum. The analysis then distinguishes failure to reject that restriction from evidence that pairwise differences in local favorable-response associations are small enough to meet a fixed practical-similarity criterion. The comparison asks how precisely an aggregate-income account can describe the association when equal contributions come from different sources. Its empirical value lies in the bounds on that simplification, including cases where the available state contexts cannot establish it.

This question concerns an evaluative step within retrospective accountability. Economic voting research connects economic conditions and assessments to incumbent support, while emphasizing that accountability depends on information and political circumstances (C. J. Anderson 2007; A. Healy and Malhotra 2013; Lewis-Beck and Stegmaier 2000). A national economic judgment is nei-

ther a direct observation of objective performance nor an electoral sanction. Respondents may draw on their own circumstances, information about others, public news, and political commitments. Studying those judgments separately makes it possible to ask whether distinct economic indicators accompany different evaluations without assuming a complete causal chain from economic conditions to presidential votes. The outcome here is the reported evaluation, with presidential approval considered only as a secondary extension.

The geographical connection is already well established as a research question. Books and Prysby relate state unemployment to retrospective national unemployment evaluations, while Ansolabehere, Meredith, and Snowberg develop an account in which information about similar others informs judgments of the aggregate economy (Ansolabehere, Meredith, and Snowberg 2014; Books and Prysby 1999). Park and Reeves examine local conditions and national perceptions across several Cooperative Congressional Election Study waves, including comparisons involving unemployment, median household income, and gasoline prices (Park and Reeves 2020, 450–51, 457–59). The present study does not introduce local context to economic voting. It asks a more specific question about the information preserved when separate accounting components are replaced by their sum.

Research on the sources of personal financial change supplies a second important precedent. Tilley, Neundorff, and Hobolt find different relationships between benefit-related and earnings-related financial changes and party support in a British panel (Tilley, Neundorff, and Hobolt 2018, 558–59, 562–63). Their individual reports allow an interpretation of financial sources that aggregate accounts cannot recover. Here, state net earnings, property-income flows, and transfers are measured on the same denominator: each component’s change relative to prior total personal income. This

common scale makes equal contributions to total growth comparable. It sacrifices the household-level evidence needed to establish which respondents received the income, recognized its source, or credited a government for it.

I apply that comparison to the 2012 and 2020 pre-election cross-sections. These years were selected in the original study proposal because the incumbent president sought reelection and are retained here as separate applications. Their observed income-composition contrast is not an ex ante selection rule. State income accounts and unemployment series describe the contexts in which respondents evaluated the national economy. Separate weighted ordinal models retain the ordered response categories. Local probability derivatives summarize the association with a favorable judgment: they are slopes of its modeled probability near respondents' observed characteristics. Direct component contrasts quantify how much those slopes could differ, rather than letting a nonsignificant common-coefficient test stand in for their precision.

The findings make that distinction consequential. The adjusted evidence does not establish the joint expectation that labor improvements and earnings growth have positive associations in both elections. Neither election rejects the common-coefficient restriction after adjustment, yet most pairwise practical-similarity comparisons remain unresolved. Two primary comparisons in 2020 satisfy the fixed criterion for small differences at the favorable-response threshold. One of these becomes unresolved when uncertainty is estimated through state deletion. These results do not establish that the components generally have different associations, but they also do not justify treating the absence of rejection as an endorsement of a sufficient total-income description.

The comparison links the accounting content of an economic indicator to the evaluative claim made with it. Reporting exact restrictions, practical margins and state-information sensitivity sepa-

rately identifies how far the available contexts support the simplification embodied in total income. This extends contextual and income-source research by making aggregation an explicit comparison within a common accounting framework. The two elections provide applications of that comparison rather than an identified pandemic or relief-policy effect; many differences between these settings remain inseparable in this design.

Table 1 describes the samples and measures; Table 2 compares the total-income and component specifications. Tables 3 and 4 trace covariate adjustment, and Tables 5 and 6 report sensitivity analyses. Table 7 summarizes the hypothesis comparisons. Tables 8 and 9 document sample construction and statistical support.

Theory and Expectations

National judgments and contextual information

The object of an economic judgment does not uniquely identify the information or motive behind it. The distinction between personal and national economic assessments is central to sociotropic voting research (Kinder and Kiewiet 1981). National assessments also figure prominently in Kinder's analysis of presidential approval across the Nixon, Ford, and Carter presidencies (Kinder 1981). Yet a national answer need not express a purely collective concern. Kramer shows why evidence commonly interpreted as sociotropic behavior can remain compatible with self-interest when personal income changes contain variation that is not politically informative (Kramer 1983). The distinction matters here because respondents evaluate the nation while the explanatory indicators describe their state. Neither wording nor geography establishes whether an answer reflects personal experience, social observation, public information, or partisan interpretation. An account of

national evaluations must leave room for these different routes to the same reported judgment.

Contextual learning offers one reason to expect geography to matter. People need not observe every part of the national economy to form a national assessment. Conditions encountered among acquaintances or similar others may provide relevant information, as Ansolabehere, Meredith, and Snowberg's incomplete-information account proposes (Ansolabehere, Meredith, and Snowberg 2014). Books and Prysby's state-unemployment analysis provides a related empirical precedent (Books and Prysby 1999). Under this account, work opportunities and income changes around a respondent may be associated with an assessment whose stated object is national. A state boundary, however, is an administrative way to summarize context rather than a demonstrated boundary of interpersonal information. Residents can have different contacts and occupations, obtain news from outside the state, and experience different local conditions within it. The geographical association therefore remains a comparison of contexts, with labor and earnings conditions providing observable implications of the information account.

Measurement and attention qualify that connection. Healy and Lenz report different conclusions about local economic accountability when population-based measures replace noisier sample-based indicators (A. Healy and Lenz 2017). Better measurement of an economic condition is valuable, but it does not establish that residents notice the condition as measured. Nadeau and colleagues describe news as an interpretive stage between elite economic assessments and public expectations; De Boef and Kellstedt likewise situate consumer sentiment within political as well as economic circumstances (De Boef and Kellstedt 2004; Nadeau et al. 1999). The distinction is between a condition that could supply information and a message that a person actually receives and accepts. An aggregate income series can improve the description of the former while leaving the latter

unobserved. Its association with a survey judgment can therefore be compatible with learning without measuring learning itself; the empirical question is which measured conditions accompany the judgment.

Political interpretation provides a competing explanation for reported judgments. Bartels's panel analysis emphasizes party attachments in political perceptions, and Evans and Pickup examine the conditioning of economic perceptions by political preferences during the 2000–2004 presidential cycle (Bartels 2002; Evans and Pickup 2010). These accounts suggest that people encountering similar conditions need not report similar evaluations. A further distinction concerns whether reported disagreement corresponds to sincere belief. Bullock and Lenz review several explanations for partisan differences in factual survey responses and caution against treating their relative importance as settled (Bullock and Lenz 2019). Party identification can therefore be relevant to an economic report without identifying a single process of motivated reasoning. Conditioning on political orientations describes an association among respondents with comparable reported orientations; it cannot isolate unbiased perception or establish the proportion of an economic association attributable to partisan motives.

Information and partisan interpretation also need not be mutually exclusive. Lewis-Beck, Martini, and Kiewiet characterize partisan bias as comparatively minor in an analysis that relates economic perceptions to gross domestic product and unemployment (Lewis-Beck, Martini, and Kiewiet 2013). That interpretation contrasts with stronger political-conditioning accounts. Dickerson and Ondercin explicitly consider how local circumstances may condition partisan motivations, while Bisgaard's British crisis study distinguishes agreement about worsening conditions from disagreement about governmental responsibility (Bisgaard 2015; Dickerson and Ondercin 2017).

Linked Swedish administrative income and survey evidence also indicates roles for economic information and partisan bias (A. J. Healy, Persson, and Snowberg 2017). These studies motivate rival interpretations rather than an assumption that political adjustment resolves them. A national economic answer can contain meaningful information while remaining politically conditioned, and an account of its contextual correlates must permit both possibilities.

Income sources and the meaning of improvement

Income composition makes the content of the economic signal more explicit. The Bureau of Economic Analysis (BEA) defines personal income to include receipts related to production as well as transfers. Its state accounts distinguish net earnings by residence, property income from dividends, interest and rent, and current transfer receipts (U.S. Bureau of Economic Analysis 2020, n.d.-b). Transfers include broad government social benefits and net current receipts from business, rather than emergency cash assistance alone (U.S. Bureau of Economic Analysis n.d.-a). These categories supply an accounting comparison with a common denominator. If a state previously had total income of 100 units, an additional unit of earnings and an additional unit of transfers each account for one percentage point of total growth. The arithmetic does not imply that the same residents receive the income or attach the same meaning to it. It makes the economic scale comparable while leaving its experiential and political interpretation open.

Labor conditions and earnings have a relatively direct connection to the contextual-learning account. A decline in unemployment or a rise in earnings may accompany more favorable work opportunities among respondents and those they observe. If those experiences inform assessments of broader economic performance, favorable labor and earnings changes should be associated with

favorable national evaluations. The unemployment precedents and incomplete-information framework motivate this expectation, rather than establishing it for every income component or electoral setting (Ansola-behere, Meredith, and Snowberg 2014; Books and Prysby 1999). Earnings also cannot be treated as a measure of market activity wholly separate from policy. BEA's discussion of the 2020 accounts includes Paycheck Protection Program support to proprietors within earnings (U.S. Bureau of Economic Analysis 2020). The expectation concerns the experiences accompanying labor and earnings conditions, not a clean distinction between income earned in markets and income sustained by government. The first hypothesis (H1) is that more favorable labor conditions and net-earnings changes are associated with more favorable national economic evaluations within each election. H1 deliberately requires this directional pattern for both labor and earnings in each of the two retained applications. That joint target is more demanding than evidence that some contextual information accompanies national judgments; failure to establish it cannot by itself distinguish the information account from its political or compositional rivals.

Transfers admit competing interpretations. They may sustain resources during hardship and may make a government response visible. Their growth can also accompany deteriorating underlying conditions. Studies of particular programs show why both the support and its attribution deserve attention. Healy and Malhotra distinguish electoral rewards for disaster relief from preparedness, while Kriner and Reeves relate the political consequences of local federal spending to responsibility and ideology (A. Healy and Malhotra 2009; Kriner and Reeves 2012). Margalit examines worker assistance as a qualification to the electoral response to trade displacement (Margalit 2011). These targeted settings cannot supply an unconditional prediction for broad state transfer growth. The same aggregate change can reflect both compensation and the circumstances requiring compensa-

tion. A negative association would not establish that recipients failed to value support; a positive association would not establish that they credited the incumbent for providing it.

The distinction between receipt and its interpretation is especially apparent in individual-level income-source research. In Tilley, Neundorff, and Hobolt's British panel, reported benefit-related financial changes generally have stronger relationships with changes in party support than earnings-related changes (Tilley, Neundorff, and Hobolt 2018, 558–59, 562–63). Their reported source of financial change contains information absent from a state account. Lacy offers a complementary caution: perceived family benefits from the federal budget do not correspond to a general beneficiary-versus-contributor vote divide after controls for most voters, with seniors an exception (Lacy 2014). Actual receipt, perceived receipt, attribution and electoral response are thus distinct constructs. Attaching a state transfer series to respondents does not collapse those distinctions. Its role is to describe the context of resources while preserving uncertainty about recipients and political meaning. That uncertainty motivates a nondirectional comparison of the transfer association with the other income sources.

Property income raises a related boundary. Work on the multidimensional economic voter distinguishes retrospective evaluations from policy positions and patrimony, including asset ownership (Lewis-Beck, Nadeau, and Foucault 2013). Dividends, interest and rent in a state's income account are flows, not measures of a respondent's assets or changes in wealth. Their association with national evaluations might reflect receipts among some residents, broader impressions of economic performance, or the state's economic composition. The aggregate cannot identify the distribution of those flows across the people answering the survey. There is consequently no defensible unconditional ordering in which the property-income association must exceed or fall below the earnings

or transfer association. Retaining property income as a separate component permits that distinction to be examined without importing an individual asset-voting explanation into a contextual measure. Responsibility further separates an economic judgment from a political reward. Anderson's multilevel-governance analysis and Ebeid and Rodden's account of state economic structure explain why assigning governmental responsibility need not be straightforward (C. D. Anderson 2006; Ebeid and Rodden 2006). Hansford and Gomez's instrumental-variable analysis also cautions against reading a contemporaneous relationship between perceptions and votes as a causal retrospective-voting test (Hansford and Gomez 2015). These considerations support treating national evaluations as a distinct outcome. Approval can indicate whether a contextual pattern appears in an adjacent political judgment, but it cannot recover a missing information or attribution pathway. Nor can an association with approval establish that an economic circumstance determined a vote. The narrower outcome is a deliberate limit on the theoretical claim, rather than a surrogate for the complete accountability process.

A common-association benchmark

The component comparison asks what is lost by representing these different sources with total income alone. Park and Reeves already compare multiple local economic indicators, while the pandemic setting has prior coverage in Sircar's work on unemployment and electoral returns and Wu and Huber's study of individual job loss (Park and Reeves 2020; Sircar 2025; Wu and Huber 2021). The present comparison instead concerns components that add to the same total and are expressed relative to the same prior income. A common-association account treats equal contributions from earnings, property income and transfers alike, conditional on the other characteristics consid-

ered. Distinct associations would be inconsistent with that restricted description, but would remain compatible with differences in attention, recipients, attribution, omitted context or measurement.

A direct contrast between component associations is therefore more informative than comparing whether each separately meets a significance threshold. The contrast asks how much the associations differ on a shared scale. Its uncertainty matters as much as its central value. Sufficiently precise differences confined within a small prespecified range would support practical similarity for the particular judgment being summarized, without establishing exact equality (Lakens 2017). An interval extending beyond that range would leave the comparison unresolved unless it lay wholly beyond the margin in a particular direction. This distinction protects the substantive question from an otherwise tempting inference: that an imprecise comparison endorses the simpler aggregate description merely because a test does not reject it. It also makes explicit that the meaning of a negligible difference requires a substantive judgment, rather than being supplied by statistical non-significance. H2 is a prespecified comparison rather than a directional hypothesis: it evaluates the common-association benchmark for net earnings, property income, and transfers using a shared lagged-total-income scale, direct component contrasts, and an equivalence interval fixed before the current model fits after earlier outcome access.

Together, H1 and H2 distinguish a directional expectation about labor and earnings from a disciplined comparison among income sources. The first follows from favorable experiences supplying information about national conditions. The second allows meaningful differences, sufficiently precise similarity, or unresolved uncertainty to inform the adequacy of an aggregate description. A common association can be positive, negative or zero, so practical similarity among components does not corroborate the positive expectation in H1. These possibilities cannot identify the under-

lying information mechanism on their own. A contextual comparison requires conditions under which the economic measures vary sufficiently and correspond to experiences relevant to respondents, while rival political and compositional explanations remain plausible. The two elections supply separate settings for this question. Their comparison does not isolate the pandemic, incumbent party or polarization, and no difference in electoral circumstances removes the need to distinguish state accounts from individual interpretation.

Data and Methods

Data and sample

The analysis uses the annual common-content releases of the 2012 and 2020 Cooperative Congressional Election Studies, subsequently named the Cooperative Election Study (Ansolabehere and Schaffner 2013; Schaffner, Ansolabehere, and Luks 2021). These are separate pre-election cross-sections. The original source samples contain 54,535 and 61,000 records, respectively. The analytic population is restricted to eligible adult respondents in the 50 states with positive finite published weights, resolved pre-election timing and complete economic linkage. Documented interview-start coverage is October 1–November 5 in 2012 and September 29–November 2 in 2020; these are start-date ranges rather than interview-completion windows (Kuriwaki 2018, 10–11). The available 2012 source documentation does not establish the timestamps’ time zone. The final analytic sample for complete-response economic models contains 53,661 respondents in 2012 and 58,272 in 2020. Approval uses separately constructed samples of 53,152 and 59,277 respondents, without requiring an economic-item response.

I apply the revised adult pre-election weight in 2012 and the common-content adult weight in 2020.

The official FAQ recommends the earlier revised weights for analyses of voters and nonvoters, and the author’s pinned cumulative-file mapping identifies the corresponding 2012 native field (Kuriwaki 2026; Schaffner and Kuriwaki n.d.). No voter-file match, turnout restriction or post-election participation requirement is imposed. The revised 2012 calibration algorithm and trimming threshold are not established by the available documentation; the original weight’s trimming rule is not attributed to the revision. These are calibration weights for an online nonprobability sample. Their use does not remove the assumptions required for population inference from such samples (Baker et al. 2013). Weights enter estimation as published and are normalized only when averaging predicted quantities.

Sample construction distinguishes eligibility, a substantive outcome response, demographic completeness and political-item completeness. A missing birth year is not automatically contradictory evidence of adult eligibility, but age must be available for a complete-control fit. Valid party or ideology responses indicating uncertainty remain categories; unanswered items do not. All principal economic models use the same complete-response sample within election. Additional models widen the sample when political or demographic controls are omitted. Table 1 reports the outcome-specific samples and modeled-variable distributions; Table 8 provides the more detailed accounting. Comparisons with excluded respondents and missing-item prevalence bounds describe the restriction rather than assume that missingness is ignorable. Each fitted sample retains all 50 states, so contextual information remains state-level despite the much larger respondent totals.

The missing-item bounds describe the prevalence of an observed favorable economic evaluation among eligible respondents. The lower bound divides observed favorable-response weight by eligible weight; the upper bound adds genuinely unanswered-item weight to the numerator. Observed

uncertainty remains in the denominator and is not reassigned as a favorable answer. These bounds do not cover regression slopes, latent beliefs or a hypothetical forced-choice response.

Variables and measures

Dependent variables The primary national economic evaluation is an ordinal variable with five substantive categories from much worse to much better. Higher values denote a more favorable assessment. The primary estimator retains all five categories; the favorable economic evaluation summary is binary and combines the two better categories. Uncertainty responses are observed answers outside this substantive ordering and are excluded from fitting, rather than treated as neutral or negative. Presidential job approval is a separate four-category outcome ordered from strong disapproval to strong approval, with the two approval categories defining its favorable summary. Its uncertainty response is handled separately on its own denominator. Similar category labels across elections do not establish measurement invariance. The analysis concerns reported judgments at interview, not within-person changes in attitudes between elections.

Independent variables State income measures come from BEA's quarterly personal-income accounts. Net earnings by residence, dividends/interest/rent, and current transfer receipts exhaust the relevant total under the accounting definitions (U.S. Bureau of Economic Analysis 2020, n.d.-a). I convert each series to real per-capita values using the same quarterly population and national personal-consumption-expenditures price index, preserving the seasonally adjusted annual-rate basis. The net-earnings contribution, property-income contribution and transfer-income contribution express each component's change relative to lagged total real per-capita income; their sum is total personal-income growth. Let the real per-capita value of component j be denoted by r and total

personal income by R . The contribution of a component to annual total-income growth is

$$C_{jst} = 100 \frac{r_{jst} - r_{js,t-4}}{R_{s,t-4}}.$$

The window runs from the second quarter of the prior year to the second quarter of the election year. One unit is a percentage-point contribution to that total, rather than one percent growth in a component or an equal dollar amount across states. A common national deflator preserves additivity but does not account for state price-level differences. The accounts use the revised September 2026 release vintage acquired on October 1, 2026: they reconstruct historical conditions rather than the exact information available to respondents. Transfers remain broader than cash relief, earnings broader than wages, and property income a flow rather than wealth. These definitions prevent a contextual comparison from being mistaken for individual benefit receipt or asset ownership.

Labor-market improvement is the prior-year minus election-year average of April–June seasonally adjusted statewide unemployment rates from the Bureau of Labor Statistics (BLS). A positive value denotes declining unemployment, expressed in percentage points. The arithmetic mean of monthly rates aligns its primary window with the income comparison. August labor-market improvement uses the corresponding August unemployment difference, leaving income at the second quarter. The prior unemployment rate is prior second-quarter unemployment, and the prior income level is log real per-capita total income in that quarter; each is centered at its unweighted state mean within election. None of these measures reproduces the complete fall-to-fall experience contemplated by a retrospective question. Nor does the availability of a release before interviewing establish that an individual encountered it.

Control variables Demographic controls include approximate age in decades centered at 45 and age squared on that centered scale. Recorded gender is binary in the source, educational attainment has six categories, and recorded race and ethnicity has eight categories. The reference groups are male, high-school graduate and White; the recorded gender measure does not exhaust gender identity. Political controls comprise five party identification categories and six political ideology categories, including valid uncertainty responses, with Independent and Moderate as reference groups. The main model conditions on these contemporary orientations; a same-sample comparison omits them. This is a comparison of conditional associations, not a causal direct-effect or mediation analysis. Industry composition, media exposure, policy choices, pandemic intensity and historical geography remain plausible omitted context. State fixed effects are not included because they would absorb the state-level exposures within each separately estimated election.

Analytic strategy

I fit separate survey-weighted cumulative-logit models with state-cluster sandwich standard errors (Lumley 2026). The primary fits use complete cases for the substantive outcome and all declared controls within each election; approval uses its own outcome-specific complete-case sample. Their proportional-odds specification uses the same slopes across the ordered-response cutpoints; this is a modeling assumption, not a property established by numerical convergence. For the economic evaluation, the main specification is

$$\begin{aligned}\text{logit Pr}(Y_{is} \leq k) &= \zeta_k - (\beta_L L_s + \beta_E C_{Es} \\ &\quad + \beta_P C_{Ps} + \beta_T C_{Ts} + B_s \gamma + D_i \delta + P_i \pi), \\ k &= 1, \dots, 4.\end{aligned}$$

The contextual, demographic and political blocks are B, D and P. Positive slopes indicate more favorable evaluations. A restricted benchmark replaces the three components with total income, imposing a common component coefficient. Total and all components never appear together in an unrestricted specification. Full coefficients and thresholds remain in the tables, while separate local derivatives summarize probability associations. This distinction matters because a logit coefficient is not a probability change, and the same coefficient can imply different local probability associations at different points of the response distribution. Pseudo-R-squared is a descriptive weighted log-likelihood improvement over the same-sample empirical-marginal model, not ordinary variance explained.

For a focal predictor, I average the derivative of the favorable-response probability over the observed election-specific sample using the corresponding weights. Here, local refers to a small change in a predictor, not a smaller geographic unit. Its uncertainty uses the full covariance of slopes and thresholds through the delta method; transformed quantities and covariance-based inference are discussed by Arel-Bundock, Greifer, and Heiss (Arel-Bundock, Greifer, and Heiss 2024). A direct component contrast subtracts two average derivatives. It describes the local direction of increasing one contribution while decreasing another by the same amount, holding total income growth fixed. This is an infinitesimal composition comparison, not a finite policy change. The approval summary uses its own threshold and observed approval sample. All inference uses a

state-cluster sandwich, with a working t distribution on 49 degrees of freedom for scalar quantities and an F distribution on 2 and 49 degrees of freedom for the joint restriction. No extra finite-cluster multiplier is applied. Across-state independence and adequate cluster information remain assumptions; cluster-robust methods do not make limited or unequal clusters inconsequential (Cameron and Miller 2015).

Each inferential family groups planned comparisons whose uncertainty is assessed together to limit false conclusions from examining multiple comparisons. This multiplicity adjustment is distinct from conditioning on demographic or political covariates. H1 has a family of four labor and earnings derivatives across the two primary models. All four lower bounds of two-sided 98.75% Bonferroni intervals must exceed zero to establish the joint positive expectation.

H2 first tests the two linear restrictions defining common component coefficients within each election; Holm adjustment applies to these two tests (Holm 1979). A separate family contains the six pairwise component-derivative differences, assessed with 99.167% intervals against a fixed margin of ± 0.020 probability units per contribution point. Strict containment establishes practical similarity for that pair; an interval wholly beyond a margin establishes a meaningful difference; otherwise the comparison is unresolved. The margin is a fixed operational tolerance of two probability percentage points in the difference between local slopes per contribution point. Before the current fits, but after earlier outcome access, the study treated a local slope difference of two favorable-probability percentage points per contribution point as large enough to change the descriptive reading of an aggregate signal. This was a study judgment. It has no externally validated basis as an electoral-indifference threshold. A common slope unit permits comparison across components but does not imply equal observed support for a one-unit change. Nonsignificance and equivalence

remain distinct (Lakens 2017). Approval’s eight focal derivatives form a separate Holm family; no article-wide familywise-error claim is made.

Sensitivity analyses address particular weaknesses rather than offer substitute opportunities to confirm H1. A binary favorable-response model examines dependence on the proportional-odds specification at that threshold. The August labor measure addresses one timing choice. Uniform fitting and averaging weights examine weighting dependence on the same respondents. Wider outcome-valid and demographic-complete samples separate completion restrictions from adjustment. Finally, deleting each state and refitting the primary component model supplies a complete state-deletion jackknife covariance; quantities are still averaged over the original 50-state grid to preserve the target of comparison.

To make the tolerance choice inspectable, I also report a post-estimation critical-tolerance summary: the larger absolute endpoint of each existing adjusted interval. Strict containment requires a tolerance exceeding that value. This arithmetic restates the same intervals without refitting models, choosing a replacement margin or revising the primary classifications.

Support and leverage diagnostics assess the information available for the component contrasts. Weight effective sample sizes do not replace the number of state contexts. Score-norm concentration depends on coefficient and threshold units and is therefore a descriptive diagnostic rather than an invariant measure of inferential information. Numerical convergence, full rank and overlap checks do not establish proportional odds or finite-cluster coverage. The cumulative-response overlap checks are not a theorem guaranteeing existence of an ordinal maximum-likelihood estimate. Hypothetical planning-power scenarios are not estimates of achieved power for the planned comparison families.

The plan preceded these fits but followed access to earlier work and outcomes; it was not externally preregistered or outcome-naïve. After estimation, the interpretation was narrowed to a common-scale comparison with mostly unresolved component contrasts.

Results

What economic circumstances lie behind total income growth?

The income decomposition distinguishes resource growth from the circumstances supplying it. Table 1 presents sample accounting and the weighted distributions of the modeled variables, while Figure 1 shows the state economic contexts. In the 2020 primary sample, total-income growth averages 9.54 percentage points, comprising -2.96 contribution points from earnings, -0.42 from property income and 12.91 from transfers. The separately rounded components sum to 9.53 . Thus a positive resource total accompanies negative earnings and property contributions, a contrast concealed by the total alone. Earnings and property contributions are positive on average in 2012, while transfers are negative. These accounting patterns do not establish how respondents evaluated any source of income.

The evaluations and labor contexts also differ across the separate settings. Favorable national evaluations account for 32.56% of weighted primary responses in 2012 and 22.80% in 2020, while weighted labor improvement averages 0.86 and -9.13 percentage points. These are descriptive differences, not an estimated pandemic effect. Table 8 shows why completion among substantive respondents differs from coverage of all eligible respondents: in 2020, 99.95% of substantive-response weight survives control completion, whereas the complete primary sample retains 94.46% of eligible weight and uncertainty accounts for 5.28% of eligible weight.

Are labor and earnings associations jointly positive?

The primary earnings estimates do not establish the positive pattern expected by H1. Table 2 reports the full total-income and component specifications; Figure 2 presents their primary local probability comparisons. Table 7A reports an earnings derivative of 0.003 (SE = 0.002), with a multiplicity-adjusted interval of $[-0.002, 0.009]$, in 2012. The corresponding 2020 derivative is -0.005 (SE = 0.002), with an adjusted interval of $[-0.012, 0.001]$. Both intervals include zero. Throughout the numerical summaries, estimates use three decimal places and sign descriptions follow unrounded values; a positive value displayed as 0.000 is not exactly zero.

The primary 2012 labor derivative is positive, with an estimate of 0.014 (SE = 0.006, $p = 0.019$, $N = 53,661$). This is a local slope of 1.4 probability percentage points per percentage point of unemployment improvement, not the estimated consequence of a finite one-point intervention. The primary 2020 labor derivative is also positive, with an estimate of 0.002 (SE = 0.002, $p = 0.360$, $N = 58,272$). The adjusted labor intervals are $[-0.001, 0.030]$ and $[-0.003, 0.006]$, respectively. The displayed p values are ordinary two-sided probabilities, whereas H1 uses the four-comparison adjustment. The ordinary significance of the earlier labor estimate or the later negative earnings estimate does not establish that all four primary associations are positive. Neither does the failure to establish H1 imply four zero associations.

Do equal income contributions have similar associations?

The restricted total-income models summarize the same respondents under the common-component restriction. The 2012 total-income model has a positive labor estimate of 0.014 (SE = 0.006, $p = 0.017$, $N = 53,661$). The 2020 total-income model has a positive labor estimate of 0.000 (SE = 0.002, $p = 0.774$, $N = 58,272$). Similar labor estimates across restricted and component speci-

cations do not test whether the income components can be combined. Table 7 reports the joint restrictions from the unrestricted models. Their ordinary p values are 0.703 and 0.480, with Holm-adjusted $p = 0.960$ for each election. Nonrejection leaves open whether differences are small or insufficiently precise.

The direct contrasts answer the practical-similarity part of H2. Every 2012 interval in Table 7 extends beyond the fixed ± 0.020 margin, leaving all three comparisons unresolved. In 2020 the earnings-property interval, $[-0.023, 0.016]$, is also unresolved. The earnings-transfer interval, $[-0.011, 0.004]$, and the property-transfer interval, $[-0.017, 0.017]$, lie strictly inside the margin under the primary uncertainty estimate. These are margin-dependent operational classifications of local favorable-probability slopes, without an externally validated claim of electoral negligibility. No interval lies wholly beyond a margin, and no election establishes all-three practical similarity. Similar slopes can also be negative or zero; similarity does not support H1's positivity claim or imply equivalence of the entire ordered response distribution.

The state-deletion comparison qualifies this limited positive evidence. Table 9 summarizes support and influence diagnostics, including variance inflation factors (VIFs), which describe collinearity in the economic design. The designs have full rank, but residual variation and leverage differ. One transfer contribution point equals 4.51 respondent-residual standard deviations in 2012 and 0.69 in 2020, as Table 9C reports. For earnings and property income, the corresponding ratios are 0.94 and 2.05 in 2012, and 1.01 and 2.51 in 2020. The fixed margin therefore places a common tolerance on slope differences despite different empirical support; these ratios neither justify a finite policy interpretation nor validate the tolerance. Six planned support warnings remain relevant. All 50 state-deletion fits succeed in each election and retain the original averaging grid. Their covariance

widens the 2020 property-transfer interval to $[-0.024, 0.024]$, leaving it unresolved, while earnings-transfer remains inside the margins at $[-0.013, 0.006]$. Table 7 retains both classifications. A large respondent sample cannot replace information across the relatively small set of economic contexts. Expressed as probability percentage points in the local slope per contribution point (100 times probability units), the 2020 critical tolerances are 1.119 for earnings-transfer, 1.743 for property-transfer and 2.251 for earnings-property; under state deletion they are 1.264, 2.426 and 2.858. These are rounded endpoints, not proposed replacement margins. Every 2012 critical tolerance exceeds two points under both calculations. At an illustrative one-point tolerance, none of the six comparisons supports similarity; at 1.5 points, only the 2020 earnings-transfer comparison does so under both calculations. The retained earnings-transfer bound therefore supports a narrower descriptive approximation than the other pairs, while leaving differences consequential to a one-point interpretation unresolved.

How does political conditioning change the comparison?

The component coefficients describe both stable directions and sensitivity to adjustment. Table 3 displays the 2012 adjustment comparisons and secondary approval model, while Table 4 presents their 2020 counterparts. Across the three economic component models in Table 3, earnings coefficients remain positive and property coefficients negative; the transfer coefficient changes from positive to negative with full adjustment. In Table 4, earnings and transfer coefficients remain negative, while the property coefficient changes sign. These descriptive coefficient patterns neither establish adjusted-significant component associations nor replace the direct derivative contrasts.

Political conditioning also accompanies a sign change in the 2012 labor association on the same respondents. The baseline-context 2012 labor derivative is negative, with an estimate of -0.006 (SE

= 0.010, $p = 0.526$, $N = 53,661$). With demographic adjustment, the 2012 labor derivative remains negative, with an estimate of -0.004 ($SE = 0.010$, $p = 0.715$, $N = 53,661$). The fully adjusted primary estimate is positive. The baseline-context 2020 derivative is positive, with an estimate of 0.003 ($SE = 0.002$, $p = 0.269$, $N = 58,272$). The demographic-adjusted 2020 derivative is positive, with an estimate of 0.003 ($SE = 0.002$, $p = 0.179$, $N = 58,272$). These conditional descriptions do not estimate a mediated share: both slopes and the nonlinear probability multiplier can change when political controls enter.

Do outcome form and timing alter the account?

Table 5 reports the complete 2012 sensitivity specifications, and Table 6 reports the corresponding 2020 specifications. Direct contrasts locate sensitivity in the component account. In 2020, the earnings–transfer difference is -0.003 in the binary model (ordinary 95% interval $[-0.008, 0.003]$) and -0.002 with uniform weights ($[-0.007, 0.002]$). The binary property–transfer interval, expressed in probability percentage points per contribution point, is $[-2.036, 1.634]$, extending below the original two-point tolerance. These ordinary intervals describe the sensitivity fits and cannot supply the adjusted primary similarity decisions. Earnings coefficients remain positive across the 2012 specifications and negative across the 2020 specifications; the earlier transfer coefficients are positive rather than negative as in the primary model. The component comparison consequently requires both the direct interval and its specification, rather than agreement in one labor slope.

The binary outcome check addresses the selected favorable threshold. The binary-outcome 2012 labor derivative is positive, with an estimate of 0.013 ($SE = 0.007$, $p = 0.086$, $N = 53,661$). The binary-outcome 2020 labor derivative is positive, with an estimate of 0.001 ($SE = 0.002$, $p = 0.672$, $N = 58,272$). The earlier labor estimate is close to its ordinal counterpart with wider ordinary

uncertainty, while the 2012 transfer point estimate changes sign. Agreement about one labor slope therefore does not validate proportional odds or all component associations.

The alternative labor window addresses timing while keeping income measured in the second quarter. The August-timing 2012 labor derivative is positive, with an estimate of 0.019 (SE = 0.008, $p = 0.018$, $N = 53,661$). The August-timing 2020 labor derivative is negative, with an estimate of -0.001 (SE = 0.003, $p = 0.619$, $N = 58,272$). The latter sign change accompanies wide uncertainty under both labor windows; it does not supply evidence that the association changed between them.

What depends on weights and response completion?

Uniform weights change both the fitted association and the distribution used for averaging. The uniformly weighted 2012 labor derivative is positive, with an estimate of 0.016 (SE = 0.005, $p = 0.002$, $N = 53,661$). The uniformly weighted 2020 labor derivative is positive, with an estimate of 0.000 (SE = 0.002, $p = 0.754$, $N = 58,272$). This alternative description is not a correction for unidentified selection or a substitute for the primary comparison.

Completion comparisons remain close to their corresponding narrower-sample descriptions. The wider demographic-complete 2012 derivative is negative, with an estimate of -0.004 (SE = 0.010, $p = 0.717$, $N = 53,730$). The wider outcome-valid 2012 derivative is negative, with an estimate of -0.006 (SE = 0.010, $p = 0.528$, $N = 53,730$). The wider demographic-complete 2020 derivative is positive, with an estimate of 0.003 (SE = 0.002, $p = 0.180$, $N = 58,305$). The wider outcome-valid 2020 derivative is positive, with an estimate of 0.003 (SE = 0.002, $p = 0.269$, $N = 58,305$). This stability concerns control-item completion among substantive respondents; it cannot establish that uncertainty responses or survey participation are ignorable.

Does the pattern extend to presidential approval?

The secondary approval outcome provides no multiplicity-adjusted extension of the contextual associations. The 2012 approval labor derivative is positive, with an estimate of 0.002 (SE = 0.006, $p = 0.695$, $N = 53,152$). The 2020 approval labor derivative is positive, with an estimate of 0.001 (SE = 0.001, $p = 0.540$, $N = 59,277$). None of the eight focal approval derivatives has a Holm-adjusted p value below 0.05; the smallest adjusted value is 0.059 for 2020 earnings. These approval-specific samples cannot be appended to the economic samples as independent observations, and approval does not measure whether an economic evaluation translated into a vote.

The income decomposition documents different circumstances within an aggregate resource change. Most comparisons of their evaluative associations remain unresolved, while limited similarity at the favorable threshold partly depends on the uncertainty calculation. Political conditioning, outcome form and state support qualify this account without selecting a learning or attribution mechanism. The evidence thus makes the aggregation assumption explicit while leaving its general adequacy unsettled.

Discussion

The component comparison places a specific limit on an aggregate-income description of national economic evaluations. In 2020, the earnings–transfer contrast is precise enough to fit within the original tolerance under both uncertainty calculations, yet not within the more demanding one-point illustrative tolerance. The property–transfer comparison requires a wider tolerance after state deletion, and the other primary comparisons remain unresolved at the study’s margin. Thus the affirmative finding concerns one pair and a stated level of descriptive precision, not interchangeable

sources of income. The evidence does not establish that the decomposed specification predicts better, or that total income should always be abandoned. It identifies how much approximation the observed component intervals permit and where an aggregate interpretation remains unsupported. Table 2, Table 7 and Figure 2 summarize the estimates and comparisons behind this interpretation. The distinction extends prior research in a specific way. Park and Reeves already connect local indicators to national perceptions, and Tilley, Neundorf, and Hobolt already distinguish the political associations of reported sources of personal financial change (Park and Reeves 2020; Tilley, Neundorf, and Hobolt 2018). The common-denominator design places contextual income flows within a single accounting comparison. It asks whether equal contributions to the same total can be summarized by similar local associations, and reports the precision with which that approximation can be supported. The surviving earnings–transfer bound supplies one answer at the observed favorable-response threshold; it neither generalizes the British panel’s individual-source finding nor tests a receipt or reward mechanism. Coherent state accounts make the aggregation restriction explicit, while sacrificing household information and finer local context. This is an incremental extension of the closest studies, with a quantitative limit on one contextual comparison rather than a new theory of economic voting.

The statistical restrictions and the behavioral accounts play different roles in this argument. The component intervals directly assess an aggregate specification. Contextual learning supplies a reason to examine state conditions, but national evaluations can also reflect news, personal experiences elsewhere, or political interpretation (Ansolabehere, Meredith, and Snowberg 2014; Evans and Pickup 2010; Nadeau et al. 1999). These processes remain observationally compatible. Adjustment sensitivity in 2012 identifies dependence on the conditioning set, not which process accounts

for it: removing political orientations does not recover an unconfounded association, and including them does not identify a direct effect. Likewise, the transfer comparison cannot separate resource maintenance, underlying distress and governmental responsibility. The study tests restrictions on contextual associations; discriminating among those individual processes requires evidence on experience and attribution.

Practical similarity remains a judgment about the precision needed for a particular description. The criterion applies to weighted average local derivatives of one favorable-response probability on the observed grid. It does not establish similar associations at every response threshold, within every subgroup, or after a finite shift in the composition of income. A small difference at the selected threshold can partly reflect respondents' positions in the ordered response distribution. The critical tolerances allow readers to inspect a stricter standard without treating it as a new primary test. Neither the retained classification nor its numerical boundary validates electoral indifference or shows that citizens use a total-income accounting rule. Similarity can also involve negative or zero common associations, so it cannot corroborate H1's positivity expectation. Lakens's distinction between nonsignificance and equivalence helps clarify that logic, but it does not supply or validate the margin selected here (Lakens 2017).

Several features of the evidence constrain stronger inference. Although each survey includes many respondents, the contextual predictors vary across only 50 states in an election. Uneven state information, high leverage and limited conditional variation can leave substantial uncertainty about differences between components. State clustering permits dependence within the specified units but still relies on across-state assumptions and a working finite-cluster calibration (Cameron and Miller 2015). Common national shocks and regional dependence can remain. The jackknife makes

influence sensitivity visible; it does not validate the coverage of every adjusted interval. These issues are especially consequential for a comparison that requires enough precision to place an entire interval inside a substantive margin. A large individual sample alone does not supply that precision at the contextual level.

Selection and measurement impose additional boundaries. Published calibration weights do not turn the underlying online nonprobability sample into a sample with known inclusion probabilities (Baker et al. 2013). Complete-response restrictions define narrower populations, while uncertainty is a substantive answer rather than an ordinary missing item. The close completion sensitivities therefore speak to one source of sample restriction, not to all selection into the survey or all willingness to evaluate the economy. State accounts also use revised economic data and a common national deflator. They omit interstate price differences, the timing of individual receipt, changing residence and the distribution of each income source among respondents. The second-quarter and August comparisons cannot reconstruct everything that a person experienced or knew before a fall interview. Similar survey response labels do not settle whether the meaning of a national economic evaluation is constant across electoral settings.

These limitations identify two distinct empirical needs. More informative contextual variation is needed to sharpen component comparisons: additional election contexts would offer more independent variation than simply enlarging the respondent sample within the same states. Finer geographic accounts could improve correspondence to local experiences where their components can be reconciled on a comparable scale. Testing the proposed mechanisms requires a different addition: linked individual records or reports of earnings and transfers, information exposure, perceived economic change and responsibility attribution. More precise contextual associations alone

would not reveal what an individual received, noticed or credited. Longitudinal observations could then distinguish changes in personal circumstances and interpretation more directly than separate cross-sections permit. The substantive objective would remain the same: determine when different sources of aggregate resources have distinguishable relationships with national judgments, while separating actual receipt, perceived receipt and political credit. That objective is not answered by interpreting uncertain aggregate comparisons as evidence that the sources do not matter.

Conclusion

Total income compresses distinct economic circumstances into a single number. That compression is convenient, but its adequacy for describing national economic evaluations is an empirical question. Comparing earnings, property income and transfers on a shared lagged-total-income scale makes the question explicit. The evidence from these two elections provides a bounded answer: the common-association restriction is not rejected, while most practical component comparisons remain uncertain and the limited similarity findings are partly sensitive to the uncertainty calculation. This distinction prevents the absence of a detected difference from being read as a general statement that income sources are interchangeable.

Different sources of income may accompany different experiences and political interpretations, but the contextual comparisons do not observe those processes. National judgments remain distinct from individual motives and electoral sanctions, and state accounts remain distinct from household experience. The immediate research priority is to separate two empirical tasks: obtaining enough contextual variation to sharpen the component bounds, and observing what individuals received, noticed and attributed to distinguish the proposed mechanisms. More respondents within the same

state contexts cannot alone accomplish the first task; more precise aggregate associations cannot alone accomplish the second. The accounting comparison provides a framework for asking both questions while keeping their evidentiary requirements separate. Its present contribution is an explicit, bounded description of what these elections permit an aggregate-income account to assume.

Data Availability

The analysis uses publicly available CCES Common Content files for 2012 and 2020, BEA state personal-income and price data, and BLS state unemployment data. The accompanying private replication materials contain code, acquisition instructions and aggregate outputs. Respondent records are not redistributed with the package. A full local rerun reproduced the numerical outputs and figure images under the documented Windows and R environment. No public deposit or release license is asserted.

AI Use Disclosure

Codex (OpenAI) assisted with literature review, research design, code development and review, local analysis, interpretation, manuscript drafting and editing, and document preparation. Respondent records were processed locally; documentation, metadata and aggregate results were used in AI-assisted review. Human review of the complete manuscript remains unconfirmed.

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Table 1. Sample accounting and weighted distributions of modeled variables**A. Sample accounting**

Characteristic	2012	2020
Eligible sample	54426	60803
Valid economic evaluation	53730	58305
Demographic-complete economic sample	53730	58305
Complete economic sample	53661	58272
Valid presidential approval	53214	59311
Demographic-complete approval sample	53214	59311
Complete approval sample	53152	59277
Economic sample: states	50	50
Economic sample: weight effective sample size	22604.91	27567.69
Approval sample: states	50	50
Approval sample: weight effective sample size	22407.84	27833.05

B. Complete economic sample

Characteristic	2012	2020
National economic evaluation: available observations	53,661 available; 0 missing	58,272 available; 0 missing
National economic evaluation: Much worse	9,332 (15.34%)	25,071 (37.74%)
National economic evaluation: Worse	13,810 (24.80%)	16,575 (29.04%)
National economic evaluation: About the same	13,016 (27.30%)	5,424 (10.43%)
National economic evaluation: Better	15,785 (29.38%)	6,225 (12.40%)
National economic evaluation: Much better	1,718 (3.17%)	4,977 (10.40%)

Characteristic	2012	2020
Presidential job approval: available observations	52,663 available; 998 missing	57,253 available; 1,019 missing
Presidential job approval: Strong disapproval	21,685 (37.70%)	30,958 (47.14%)
Presidential job approval: Some disapproval	4,292 (9.30%)	3,976 (7.13%)
Presidential job approval: Some approval	12,929 (27.99%)	8,131 (16.02%)
Presidential job approval: Strong approval	13,757 (25.01%)	14,188 (29.70%)
Favorable economic evaluation: available observations	53,661 available; 0 missing	58,272 available; 0 missing
Favorable economic evaluation: Same or worse	36,158 (67.44%)	47,070 (77.20%)
Favorable economic evaluation: Better or much better	17,503 (32.56%)	11,202 (22.80%)
Labor-market improvement: available observations	53,661 available; 0 missing	58,272 available; 0 missing
Labor-market improvement: mean (SD)	0.86 (0.61)	-9.13 (2.55)
Net-earnings contribution: available observations	53,661 available; 0 missing	58,272 available; 0 missing
Net-earnings contribution: mean (SD)	1.10 (1.19)	-2.96 (1.36)
Property-income contribution: available observations	53,661 available; 0 missing	58,272 available; 0 missing
Property-income contribution: mean (SD)	1.27 (0.54)	-0.42 (0.48)
Transfer-income contribution: available observations	53,661 available; 0 missing	58,272 available; 0 missing
Transfer-income contribution: mean (SD)	-0.42 (0.24)	12.91 (2.29)
Total personal-income growth: available observations	53,661 available; 0 missing	58,272 available; 0 missing
Total personal-income growth: mean (SD)	1.95 (1.44)	9.54 (2.44)
August labor-market improvement: available observations	53,661 available; 0 missing	58,272 available; 0 missing

Characteristic	2012	2020
August labor-market improvement: mean (SD)	1.00 (0.59)	-4.89 (1.93)
Prior unemployment rate: available observations	53,661 available; 0 missing	58,272 available; 0 missing
Prior unemployment rate: mean (SD)	0.83 (1.73)	0.11 (0.57)
Prior income level: available observations	53,661 available; 0 missing	58,272 available; 0 missing
Prior income level: mean (SD)	0.01 (0.13)	0.03 (0.14)
Age: available observations	53,661 available; 0 missing	58,272 available; 0 missing
Age: mean (SD)	0.18 (1.68)	0.39 (1.79)
Age squared: available observations	53,661 available; 0 missing	58,272 available; 0 missing
Age squared: mean (SD)	2.86 (2.73)	3.37 (3.37)
Recorded gender: available observations	53,661 available; 0 missing	58,272 available; 0 missing
Recorded gender: Male	25,311 (48.28%)	24,900 (48.72%)
Recorded gender: Female	28,350 (51.72%)	33,372 (51.28%)
Educational attainment: available observations	53,661 available; 0 missing	58,272 available; 0 missing
Educational attainment: High-school graduate	13,501 (30.61%)	15,580 (28.23%)
Educational attainment: Less than high school	1,495 (9.24%)	1,742 (8.16%)
Educational attainment: Some college	14,363 (24.53%)	12,755 (21.45%)
Educational attainment: Two-year degree	5,343 (8.90%)	6,282 (10.29%)
Educational attainment: Four-year degree	11,928 (17.46%)	13,756 (20.23%)
Educational attainment: Postgraduate education	7,031 (9.26%)	8,157 (11.64%)
Recorded race and ethnicity: available observations	53,661 available; 0 missing	58,272 available; 0 missing
Recorded race and ethnicity: White	39,897 (74.28%)	42,443 (69.63%)
Recorded race and ethnicity: Black	6,491 (11.70%)	6,477 (12.45%)
Recorded race and ethnicity: Hispanic	4,042 (7.56%)	4,871 (9.64%)

Characteristic	2012	2020
Recorded race and ethnicity: Asian	805 (2.04%)	1,745 (4.30%)
Recorded race and ethnicity: Native American	402 (0.73%)	453 (0.68%)
Recorded race and ethnicity: Mixed	1,029 (1.99%)	1,254 (1.98%)
Recorded race and ethnicity: Other	938 (1.53%)	947 (1.21%)
Recorded race and ethnicity: Middle Eastern	57 (0.17%)	82 (0.12%)
Party identification: available observations	53,661 available; 0 missing	58,272 available; 0 missing
Party identification: Independent	14,837 (32.50%)	16,419 (26.75%)
Party identification: Democrat	19,848 (32.41%)	22,114 (34.30%)
Party identification: Republican	14,421 (26.89%)	14,624 (29.00%)
Party identification: Other	2,277 (3.79%)	2,306 (3.51%)
Party identification: Not sure	2,278 (4.40%)	2,809 (6.44%)
Political ideology: available observations	53,661 available; 0 missing	58,272 available; 0 missing
Political ideology: Moderate	16,136 (33.88%)	18,534 (30.36%)
Political ideology: Very liberal	4,497 (4.87%)	7,905 (11.09%)
Political ideology: Liberal	9,473 (19.37%)	11,049 (16.35%)
Political ideology: Conservative	13,058 (28.69%)	10,328 (19.82%)
Political ideology: Very conservative	6,891 (6.64%)	6,571 (13.84%)
Political ideology: Not sure	3,606 (6.55%)	3,885 (8.54%)

C. Complete approval sample

Characteristic	2012	2020
National economic evaluation: available observations	52,663 available; 489 missing	57,253 available; 2,024 missing
National economic evaluation: Much worse	9,242 (15.50%)	24,754 (37.92%)
National economic evaluation: Worse	13,520 (24.73%)	16,225 (28.95%)

Characteristic	2012	2020
National economic evaluation: About the same	12,529 (26.82%)	5,198 (10.16%)
National economic evaluation: Better	15,661 (29.73%)	6,135 (12.47%)
National economic evaluation: Much better	1,711 (3.22%)	4,941 (10.51%)
Presidential job approval: available observations	53,152 available; 0 missing	59,277 available; 0 missing
Presidential job approval: Strong disapproval	21,740 (37.41%)	31,757 (46.67%)
Presidential job approval: Some disapproval	4,345 (9.30%)	4,193 (7.33%)
Presidential job approval: Some approval	13,121 (28.11%)	8,596 (16.36%)
Presidential job approval: Strong approval	13,946 (25.18%)	14,731 (29.64%)
Favorable economic evaluation: available observations	52,663 available; 489 missing	57,253 available; 2,024 missing
Favorable economic evaluation: Same or worse	35,291 (67.05%)	46,177 (77.03%)
Favorable economic evaluation: Better or much better	17,372 (32.95%)	11,076 (22.97%)
Labor-market improvement: available observations	53,152 available; 0 missing	59,277 available; 0 missing
Labor-market improvement: mean (SD)	0.86 (0.61)	-9.13 (2.54)
Net-earnings contribution: available observations	53,152 available; 0 missing	59,277 available; 0 missing
Net-earnings contribution: mean (SD)	1.10 (1.19)	-2.96 (1.36)
Property-income contribution: available observations	53,152 available; 0 missing	59,277 available; 0 missing
Property-income contribution: mean (SD)	1.27 (0.54)	-0.42 (0.48)
Transfer-income contribution: available observations	53,152 available; 0 missing	59,277 available; 0 missing

Characteristic	2012	2020
Transfer-income contribution: mean (SD)	-0.42 (0.24)	12.92 (2.30)
Total personal-income growth: available observations	53,152 available; 0 missing	59,277 available; 0 missing
Total personal-income growth: mean (SD)	1.95 (1.44)	9.54 (2.45)
August labor-market improvement: available observations	53,152 available; 0 missing	59,277 available; 0 missing
August labor-market improvement: mean (SD)	1.00 (0.59)	-4.89 (1.92)
Prior unemployment rate: available observations	53,152 available; 0 missing	59,277 available; 0 missing
Prior unemployment rate: mean (SD)	0.83 (1.73)	0.11 (0.57)
Prior income level: available observations	53,152 available; 0 missing	59,277 available; 0 missing
Prior income level: mean (SD)	0.01 (0.13)	0.03 (0.14)
Age: available observations	53,152 available; 0 missing	59,277 available; 0 missing
Age: mean (SD)	0.18 (1.68)	0.37 (1.80)
Age squared: available observations	53,152 available; 0 missing	59,277 available; 0 missing
Age squared: mean (SD)	2.86 (2.73)	3.39 (3.37)
Recorded gender: available observations	53,152 available; 0 missing	59,277 available; 0 missing
Recorded gender: Male	25,154 (48.28%)	25,268 (48.69%)
Recorded gender: Female	27,998 (51.72%)	34,009 (51.31%)
Educational attainment: available observations	53,152 available; 0 missing	59,277 available; 0 missing
Educational attainment: High-school graduate	13,297 (30.51%)	15,863 (28.23%)
Educational attainment: Less than high school	1,465 (9.19%)	1,811 (8.36%)
Educational attainment: Some college	14,256 (24.61%)	13,010 (21.54%)
Educational attainment: Two-year degree	5,273 (8.86%)	6,404 (10.32%)
Educational attainment: Four-year degree	11,859 (17.54%)	13,938 (20.00%)
Educational attainment: Postgraduate education	7,002 (9.29%)	8,251 (11.55%)

Characteristic	2012	2020
Recorded race and ethnicity: available observations	53,152 available; 0 missing	59,277 available; 0 missing
Recorded race and ethnicity: White	39,508 (74.16%)	43,195 (69.60%)
Recorded race and ethnicity: Black	6,511 (11.89%)	6,571 (12.48%)
Recorded race and ethnicity: Hispanic	3,959 (7.53%)	4,962 (9.63%)
Recorded race and ethnicity: Asian	760 (1.96%)	1,753 (4.22%)
Recorded race and ethnicity: Native American	401 (0.73%)	455 (0.67%)
Recorded race and ethnicity: Mixed	1,021 (2.01%)	1,282 (2.04%)
Recorded race and ethnicity: Other	936 (1.55%)	977 (1.23%)
Recorded race and ethnicity: Middle Eastern	56 (0.17%)	82 (0.12%)
Party identification: available observations	53,152 available; 0 missing	59,277 available; 0 missing
Party identification: Independent	14,662 (32.39%)	16,664 (26.72%)
Party identification: Democrat	19,860 (32.75%)	22,412 (34.06%)
Party identification: Republican	14,342 (26.97%)	15,085 (29.33%)
Party identification: Other	2,239 (3.75%)	2,331 (3.50%)
Party identification: Not sure	2,049 (4.14%)	2,785 (6.38%)
Political ideology: available observations	53,152 available; 0 missing	59,277 available; 0 missing
Political ideology: Moderate	15,996 (33.86%)	18,820 (30.30%)
Political ideology: Very liberal	4,507 (4.93%)	8,002 (11.01%)
Political ideology: Liberal	9,439 (19.50%)	11,199 (16.20%)
Political ideology: Conservative	12,988 (28.76%)	10,590 (19.92%)
Political ideology: Very conservative	6,889 (6.70%)	6,744 (13.94%)
Political ideology: Not sure	3,333 (6.25%)	3,922 (8.62%)

Note: Continuous variables report the weighted mean and standard deviation; categories report unweighted counts and weighted percentages. Each distribution uses nonmissing values of that variable within its separately defined sample. Available and missing counts show the denominator, including the other outcome measured in that sample. Economic and approval samples overlap and must not be added.

Note: Labor measures are percentage-point declines in unemployment. Income components use prior total real per-capita income as the common denominator. Prior unemployment and prior log income are centered; age is in decades centered at 45 and age squared uses that centered scale. Economic evaluation categories run from much worse to much better; approval categories run from strong disapproval to strong approval.

Note: The complete local supplementary export additionally retains ranges, weight masses and excluded-case distributions. Weight effective sample sizes summarize calibration-weight concentration and are not numbers of independent state contexts.

Table 2. National economic evaluations: total and component income models

Term	Model 1: 2012: Total income	Model 2: 2012: Components	Model 3: 2020: Total income	Model 4: 2020: Components
Labor-market improvement	0.085 (0.035)	0.087 (0.036)	0.003 (0.011)	0.011 (0.012)
Net-earnings contribution		0.021 (0.014)		-0.035 (0.017)
Property-income contribution		-0.012 (0.038)		-0.013 (0.040)
Transfer-income contribution		-0.008 (0.068)		-0.011 (0.010)
Total income growth	0.012 (0.011)		-0.019 (0.008)	
Prior unemployment	-0.010 (0.011)	-0.011 (0.011)	0.021 (0.038)	0.008 (0.038)
Prior log income	0.309 (0.172)	0.363 (0.196)	-0.429 (0.179)	-0.327 (0.198)
Age, decades centered at 45	-0.054 (0.010)	-0.054 (0.010)	-0.097 (0.012)	-0.097 (0.012)
Squared centered age	0.024 (0.006)	0.024 (0.006)	0.042 (0.005)	0.042 (0.005)
Recorded gender: female	-0.312 (0.026)	-0.312 (0.026)	-0.169 (0.016)	-0.169 (0.016)
Education: Less than high school	-0.241 (0.070)	-0.239 (0.069)	-0.102 (0.110)	-0.102 (0.110)
Education: some college	0.170 (0.031)	0.171 (0.031)	-0.075 (0.038)	-0.073 (0.038)
Education: two-year degree	0.170 (0.035)	0.170 (0.035)	-0.021 (0.044)	-0.018 (0.045)
Education: four-year degree	0.424 (0.035)	0.423 (0.035)	-0.196 (0.037)	-0.194 (0.036)
Education: postgraduate education	0.577 (0.046)	0.577 (0.046)	-0.361 (0.047)	-0.361 (0.047)
Race and ethnicity: Black	0.849 (0.079)	0.847 (0.079)	-0.062 (0.044)	-0.064 (0.045)
Race and ethnicity: Hispanic	0.058 (0.047)	0.058 (0.048)	0.203 (0.056)	0.208 (0.054)
Race and ethnicity: Asian	0.145 (0.070)	0.144 (0.069)	0.183 (0.069)	0.187 (0.068)
Race and ethnicity: Native American	-0.268 (0.192)	-0.269 (0.192)	-0.008 (0.107)	-0.004 (0.108)
Race and ethnicity: mixed in 2012 or two or more races in 2020	0.089 (0.128)	0.088 (0.128)	-0.118 (0.118)	-0.114 (0.118)
Race and ethnicity: other	-0.216 (0.119)	-0.215 (0.119)	0.274 (0.097)	0.276 (0.098)
Race and ethnicity: Middle Eastern	-0.430 (0.249)	-0.435 (0.255)	0.286 (0.160)	0.280 (0.160)
Party identification: Democrat	0.856 (0.045)	0.856 (0.045)	-0.645 (0.032)	-0.646 (0.032)

Term	Model 1: 2012: Total income	Model 2: 2012: Components	Model 3: 2020: Total income	Model 4: 2020: Components
Party identification: Republican	-0.715 (0.043)	-0.715 (0.043)	0.670 (0.053)	0.669 (0.054)
Party identification: other	-0.467 (0.088)	-0.467 (0.088)	-0.055 (0.061)	-0.055 (0.061)
Party identification: not sure	-0.063 (0.045)	-0.065 (0.045)	0.035 (0.063)	0.034 (0.063)
Political ideology: Very liberal	0.780 (0.057)	0.780 (0.057)	-1.060 (0.043)	-1.059 (0.043)
Political ideology: liberal	0.510 (0.041)	0.511 (0.041)	-0.555 (0.044)	-0.554 (0.044)
Political ideology: conservative	-1.064 (0.036)	-1.064 (0.036)	0.713 (0.043)	0.714 (0.043)
Political ideology: very conservative	-1.688 (0.060)	-1.688 (0.060)	1.101 (0.042)	1.102 (0.043)
Political ideology: not sure	-0.552 (0.043)	-0.551 (0.043)	-0.154 (0.060)	-0.154 (0.060)
Threshold 1 2	-2.251 (0.082)	-2.262 (0.100)	-0.890 (0.102)	-0.819 (0.133)
Threshold 2 3	-0.558 (0.081)	-0.569 (0.097)	0.631 (0.102)	0.703 (0.132)
Threshold 3 4	1.010 (0.086)	1.000 (0.099)	1.267 (0.101)	1.339 (0.131)
Threshold 4 5	4.109 (0.125)	4.099 (0.138)	2.343 (0.109)	2.415 (0.143)
Respondents	53661	53661	58272	58272
States	50	50	50	50
Weighted pseudo-R-squared	0.140	0.140	0.105	0.105
Fitting sample	Complete primary sample	Complete primary sample	Complete primary sample	Complete primary sample
Weight policy	published common-content weights	published common-content weights	published common-content weights	published common-content weights
Recorded gender: Male (reference)	Reference	Reference	Reference	Reference
Education: high-school graduate (reference)	Reference	Reference	Reference	Reference
Race and ethnicity: White (reference)	Reference	Reference	Reference	Reference
Party identification: independent (reference)	Reference	Reference	Reference	Reference

Term	Model 1: 2012: Total income	Model 2: 2012: Components	Model 3: 2020: Total income	Model 4: 2020: Components
Political ideology: moderate (reference)	Reference	Reference	Reference	Reference
Probability derivatives: distinct favorable-response scale				
Average local labor derivative (SE)	0.014 (0.006)	0.014 (0.006)	0.000 (0.002)	0.002 (0.002)
Ordinary labor-derivative p	0.017	0.019	0.774	0.360
Derivative sample size	53661	53661	58272	58272

Note: Entries are cumulative-logit slopes and thresholds, with state-cluster standard errors in parentheses; the binary-logit sensitivity has an intercept instead of thresholds. Higher ordered responses are more favorable. All estimated slopes, categorical levels and thresholds are retained; reference categories appear below. Threshold $k \mid k+1$ separates ordered categories 1 through k from higher categories; economic evaluations have five categories and approval has four. Income variables are percentage-point contributions relative to prior total real per-capita income; total income is percentage growth. Labor improvement is a percentage-point decline in unemployment. Published adult common-content weights are used except where uniform weights are stated. State is a working dependence cluster, not an original survey primary sampling unit. No extra finite-cluster multiplier or finite-population correction is applied. Models are separate by election; the approximation assumes independent states. The narrative uses average probability derivatives, not these logit slopes. Pseudo-R-squared is a descriptive weighted log-likelihood improvement over the same-sample empirical-marginal model, not ordinary variance explained. No significance stars are used; exact probabilities from the working t distribution with 49 degrees of freedom are retained in the complete coefficient supplement. Complete-sample restrictions differ where the table says so.

Note: The final block reports average local derivatives of the favorable-response probability and their state-cluster standard errors, in probability units per one percentage point of labor improvement. Ordinary p -values in this block use the working t distribution with 49 degrees of freedom; they do not apply the primary or secondary multiplicity adjustments. Economic favorability is category 4 or 5 in the ordinal models and the positive binary outcome in the binary model; approval favorability is category 3 or 4. The August specification uses August labor improvement. These derivatives are distinct from the logit coefficients above.

Table 3. Adjustment comparisons and approval: 2012

Term	Model 1: Baseline context	Model 2: Demographic adjustment	Model 3: Full adjustment	Model 4: Total income	Model 5: Presidential approval
Labor-market improvement	-0.028 (0.045)	-0.019 (0.051)	0.087 (0.036)	0.085 (0.035)	0.017 (0.043)
Net-earnings contribution	0.017 (0.016)	0.020 (0.019)	0.021 (0.014)		0.013 (0.018)
Property-income contribution	-0.077 (0.036)	-0.016 (0.042)	-0.012 (0.038)		-0.055 (0.045)
Transfer-income contribution	0.016 (0.083)	0.031 (0.084)	-0.008 (0.068)		-0.107 (0.090)
Total income growth				0.012 (0.011)	
Prior unemployment	0.051 (0.013)	0.033 (0.015)	-0.011 (0.011)	-0.010 (0.011)	-0.002 (0.011)
Prior log income	1.015 (0.218)	0.872 (0.244)	0.363 (0.196)	0.309 (0.172)	0.685 (0.198)
Age, decades centered at 45		-0.104 (0.008)	-0.054 (0.010)	-0.054 (0.010)	-0.047 (0.011)
Squared centered age		0.016 (0.005)	0.024 (0.006)	0.024 (0.006)	0.000 (0.006)
Recorded gender: female		-0.115 (0.026)	-0.312 (0.026)	-0.312 (0.026)	0.071 (0.030)
Education: Less than high school		-0.181 (0.071)	-0.239 (0.069)	-0.241 (0.070)	0.192 (0.099)
Education: some college		0.122 (0.029)	0.171 (0.031)	0.170 (0.031)	0.035 (0.038)
Education: two-year degree		0.121 (0.033)	0.170 (0.035)	0.170 (0.035)	0.074 (0.047)
Education: four-year degree		0.415 (0.037)	0.423 (0.035)	0.424 (0.035)	0.019 (0.041)
Education: postgraduate education		0.744 (0.047)	0.577 (0.046)	0.577 (0.046)	0.111 (0.049)
Race and ethnicity: Black		1.424 (0.076)	0.847 (0.079)	0.849 (0.079)	1.591 (0.082)
Race and ethnicity: Hispanic		0.365 (0.039)	0.058 (0.048)	0.058 (0.047)	0.563 (0.043)
Race and ethnicity: Asian		0.438 (0.068)	0.144 (0.069)	0.145 (0.070)	0.357 (0.092)
Race and ethnicity: Native American		-0.361 (0.182)	-0.269 (0.192)	-0.268 (0.192)	-0.057 (0.185)
Race and ethnicity: mixed in 2012 or two or more races in 2020		0.364 (0.130)	0.088 (0.128)	0.089 (0.128)	0.264 (0.136)
Race and ethnicity: other		-0.372 (0.154)	-0.215 (0.119)	-0.216 (0.119)	0.169 (0.124)

Term	Model 1: Baseline context	Model 2: Demographic adjustment	Model 3: Full adjustment	Model 4: Total income	Model 5: Presidential approval
Race and ethnicity: Middle Eastern		-0.277 (0.269)	-0.435 (0.255)	-0.430 (0.249)	-0.196 (0.692)
Party identification: Democrat			0.856 (0.045)	0.856 (0.045)	1.392 (0.039)
Party identification: Republican			-0.715 (0.043)	-0.715 (0.043)	-1.590 (0.067)
Party identification: other			-0.467 (0.088)	-0.467 (0.088)	-0.690 (0.103)
Party identification: not sure			-0.065 (0.045)	-0.063 (0.045)	0.053 (0.057)
Political ideology: Very liberal			0.780 (0.057)	0.780 (0.057)	0.966 (0.050)
Political ideology: liberal			0.511 (0.041)	0.510 (0.041)	0.620 (0.035)
Political ideology: conservative			-1.064 (0.036)	-1.064 (0.036)	-1.377 (0.044)
Political ideology: very conservative			-1.688 (0.060)	-1.688 (0.060)	-1.958 (0.106)
Political ideology: not sure			-0.551 (0.043)	-0.552 (0.043)	-0.142 (0.062)
Threshold 1 2	-1.773 (0.082)	-1.460 (0.109)	-2.262 (0.100)	-2.251 (0.082)	-0.919 (0.101)
Threshold 2 3	-0.459 (0.079)	-0.082 (0.108)	-0.569 (0.097)	-0.558 (0.081)	-0.149 (0.101)
Threshold 3 4	0.675 (0.082)	1.129 (0.112)	1.000 (0.099)	1.010 (0.086)	1.928 (0.106)
Threshold 4 5	3.370 (0.114)	3.931 (0.153)	4.099 (0.138)	4.109 (0.125)	
Respondents	53661	53661	53661	53661	53152
States	50	50	50	50	50
Weighted pseudo-R-squared	0.002	0.032	0.140	0.140	0.279
Fitting sample	Complete primary sample	Complete primary sample	Complete primary sample	Complete primary sample	Complete approval sample
Weight policy	published common-content weights	published common-content weights	published common-content weights	published common-content weights	published common-content weights
Recorded gender: Male (reference)		Reference	Reference	Reference	Reference

Term	Model 1: Baseline context	Model 2: Demographic adjustment	Model 3: Full adjustment	Model 4: Total income	Model 5: Presidential approval
Education: high-school graduate (reference)		Reference	Reference	Reference	Reference
Race and ethnicity: White (reference)		Reference	Reference	Reference	Reference
Party identification: independent (reference)			Reference	Reference	Reference
Political ideology: moderate (reference)			Reference	Reference	Reference
Probability derivatives: distinct favorable-response scale					
Average local labor derivative (SE)	-0.006 (0.010)	-0.004 (0.010)	0.014 (0.006)	0.014 (0.006)	0.002 (0.006)
Ordinary labor-derivative p	0.526	0.715	0.019	0.017	0.695
Derivative sample size	53661	53661	53661	53661	53152

Note: The 2012 adjustment columns use the same primary economic sample; the approval column uses its separate outcome-specific sample. The common definitions of slopes, thresholds, derivative units, reference categories, weights and working state-cluster inference are given in the notes to Table 2. Ordinary probabilities in the derivative block do not apply the primary or secondary multiplicity adjustments.

Table 4. Adjustment comparisons and approval: 2020

Term	Model 1: Baseline context	Model 2: Demographic adjustment	Model 3: Full adjustment	Model 4: Total income	Model 5: Presidential approval
Labor-market improvement	0.015 (0.013)	0.018 (0.013)	0.011 (0.012)	0.003 (0.011)	0.005 (0.008)
Net-earnings contribution	-0.045 (0.016)	-0.057 (0.017)	-0.035 (0.017)		-0.049 (0.018)
Property-income contribution	0.012 (0.044)	-0.020 (0.045)	-0.013 (0.040)		-0.086 (0.041)
Transfer-income contribution	-0.024 (0.010)	-0.033 (0.011)	-0.011 (0.010)		-0.020 (0.014)
Total income growth				-0.019 (0.008)	
Prior unemployment	-0.032 (0.038)	-0.009 (0.045)	0.008 (0.038)	0.021 (0.038)	0.039 (0.039)
Prior log income	-1.304 (0.191)	-1.328 (0.203)	-0.327 (0.198)	-0.429 (0.179)	-0.950 (0.174)
Age, decades centered at 45		-0.012 (0.014)	-0.097 (0.012)	-0.097 (0.012)	0.047 (0.012)
Squared centered age		0.037 (0.005)	0.042 (0.005)	0.042 (0.005)	-0.003 (0.004)
Recorded gender: female		-0.270 (0.016)	-0.169 (0.016)	-0.169 (0.016)	-0.222 (0.025)
Education: Less than high school		-0.119 (0.086)	-0.102 (0.110)	-0.102 (0.110)	0.110 (0.067)
Education: some college		-0.154 (0.030)	-0.073 (0.038)	-0.075 (0.038)	-0.110 (0.039)
Education: two-year degree		-0.046 (0.038)	-0.018 (0.045)	-0.021 (0.044)	-0.091 (0.041)
Education: four-year degree		-0.392 (0.040)	-0.194 (0.036)	-0.196 (0.037)	-0.467 (0.056)
Education: postgraduate education		-0.752 (0.048)	-0.361 (0.047)	-0.361 (0.047)	-0.675 (0.067)
Race and ethnicity: Black		-0.698 (0.062)	-0.064 (0.045)	-0.062 (0.044)	-0.931 (0.072)
Race and ethnicity: Hispanic		-0.182 (0.053)	0.208 (0.054)	0.203 (0.056)	-0.366 (0.055)
Race and ethnicity: Asian		0.024 (0.060)	0.187 (0.068)	0.183 (0.069)	-0.117 (0.110)
Race and ethnicity: Native American		-0.090 (0.108)	-0.004 (0.108)	-0.008 (0.107)	0.148 (0.175)
Race and ethnicity: mixed in 2012 or two or more races in 2020		-0.396 (0.128)	-0.114 (0.118)	-0.118 (0.118)	-0.446 (0.111)
Race and ethnicity: other		0.178 (0.093)	0.276 (0.098)	0.274 (0.097)	0.194 (0.128)

Term	Model 1: Baseline context	Model 2: Demographic adjustment	Model 3: Full adjustment	Model 4: Total income	Model 5: Presidential approval
Race and ethnicity: Middle Eastern		-0.058 (0.168)	0.280 (0.160)	0.286 (0.160)	0.527 (0.214)
Party identification: Democrat			-0.646 (0.032)	-0.645 (0.032)	-1.798 (0.064)
Party identification: Republican			0.669 (0.054)	0.670 (0.053)	1.267 (0.048)
Party identification: other			-0.055 (0.061)	-0.055 (0.061)	0.106 (0.064)
Party identification: not sure			0.034 (0.063)	0.035 (0.063)	-0.116 (0.070)
Political ideology: Very liberal			-1.059 (0.043)	-1.060 (0.043)	-1.761 (0.106)
Political ideology: liberal			-0.554 (0.044)	-0.555 (0.044)	-1.111 (0.065)
Political ideology: conservative			0.714 (0.043)	0.713 (0.043)	1.151 (0.036)
Political ideology: very conservative			1.102 (0.043)	1.101 (0.042)	1.999 (0.064)
Political ideology: not sure			-0.154 (0.060)	-0.154 (0.060)	0.009 (0.058)
Threshold 1 2	-0.858 (0.116)	-1.313 (0.133)	-0.819 (0.133)	-0.890 (0.102)	-0.987 (0.187)
Threshold 2 3	0.350 (0.116)	-0.064 (0.133)	0.703 (0.132)	0.631 (0.102)	-0.356 (0.189)
Threshold 3 4	0.875 (0.116)	0.475 (0.134)	1.339 (0.131)	1.267 (0.101)	1.052 (0.195)
Threshold 4 5	1.813 (0.127)	1.429 (0.144)	2.415 (0.143)	2.343 (0.109)	
Respondents	58272	58272	58272	58272	59277
States	50	50	50	50	50
Weighted pseudo-R-squared	0.003	0.018	0.105	0.105	0.325
Fitting sample	Complete	Complete	Complete	Complete	Complete
Weight policy	primary sample published	primary sample published	primary sample published	primary sample published	approval sample published
	common-content weights	common-content weights	common-content weights	common-content weights	common-content weights
Recorded gender: Male (reference)		Reference	Reference	Reference	Reference

Term	Model 1: Baseline context	Model 2: Demographic adjustment	Model 3: Full adjustment	Model 4: Total income	Model 5: Presidential approval
Education: high-school graduate (reference)		Reference	Reference	Reference	Reference
Race and ethnicity: White (reference)		Reference	Reference	Reference	Reference
Party identification: independent (reference)			Reference	Reference	Reference
Political ideology: moderate (reference)			Reference	Reference	Reference
Probability derivatives: distinct favorable-response scale					
Average local labor derivative (SE)	0.003 (0.002)	0.003 (0.002)	0.002 (0.002)	0.000 (0.002)	0.001 (0.001)
Ordinary labor-derivative p	0.269	0.179	0.360	0.774	0.540
Derivative sample size	58272	58272	58272	58272	59277

Note: The 2020 approval column need not contain the same respondents as the economic-evaluation columns. Table 2 supplies the shared coefficient, threshold and local-derivative definitions and the common statistical conventions. Political adjustment changes conditional descriptions on the common economic sample; it is not a mediation comparison.

Table 5. Planned sensitivity models: 2012

Term	Model 1: Binary favorable	Model 2: August labor	Model 3: Broader demographics	Model 4: Broader valid outcomes	Model 5: Uniform weights
Labor-market improvement	0.081 (0.047)		-0.019 (0.051)	-0.028 (0.045)	0.105 (0.032)
August labor improvement		0.113 (0.047)			
Net-earnings contribution	0.031 (0.017)	0.016 (0.013)	0.019 (0.019)	0.016 (0.016)	0.030 (0.011)
Property-income contribution	-0.062 (0.044)	-0.020 (0.036)	-0.016 (0.042)	-0.076 (0.037)	-0.017 (0.028)
Transfer-income contribution	0.066 (0.079)	0.003 (0.070)	0.032 (0.085)	0.019 (0.084)	0.056 (0.069)
Prior unemployment	-0.009 (0.014)	-0.018 (0.013)	0.033 (0.015)	0.050 (0.013)	-0.023 (0.009)
Prior log income	0.419 (0.220)	0.385 (0.194)	0.869 (0.243)	1.007 (0.217)	0.108 (0.159)
Age, decades centered at 45	0.060 (0.013)	-0.054 (0.010)	-0.105 (0.008)		-0.034 (0.009)
Squared centered age	0.042 (0.006)	0.024 (0.006)	0.016 (0.005)		0.016 (0.004)
Recorded gender: female	-0.370 (0.039)	-0.312 (0.026)	-0.114 (0.026)		-0.278 (0.015)
Education: Less than high school	-0.195 (0.092)	-0.241 (0.069)	-0.174 (0.072)		-0.141 (0.045)
Education: some college	0.313 (0.045)	0.170 (0.031)	0.123 (0.029)		0.152 (0.017)
Education: two-year degree	0.277 (0.055)	0.170 (0.035)	0.121 (0.033)		0.124 (0.027)
Education: four-year degree	0.513 (0.046)	0.423 (0.035)	0.414 (0.037)		0.394 (0.017)
Education: postgraduate education	0.676 (0.069)	0.577 (0.046)	0.744 (0.047)		0.549 (0.028)
Race and ethnicity: Black	0.786 (0.080)	0.843 (0.078)	1.420 (0.075)		0.761 (0.043)
Race and ethnicity: Hispanic	0.014 (0.046)	0.053 (0.049)	0.363 (0.038)		0.002 (0.056)
Race and ethnicity: Asian	0.179 (0.069)	0.137 (0.070)	0.437 (0.068)		0.044 (0.066)
Race and ethnicity: Native American	0.009 (0.164)	-0.272 (0.192)	-0.362 (0.182)		-0.260 (0.093)
Race and ethnicity: mixed in 2012 or two or more races in 2020	0.210 (0.141)	0.082 (0.127)	0.363 (0.130)		0.074 (0.081)
Race and ethnicity: other	0.247 (0.138)	-0.220 (0.118)	-0.372 (0.154)		-0.362 (0.075)
Race and ethnicity: Middle Eastern	-0.722 (0.499)	-0.447 (0.259)	-0.279 (0.269)		-0.392 (0.280)

Term	Model 1: Binary favorable	Model 2: August labor	Model 3: Broader demographics	Model 4: Broader valid outcomes	Model 5: Uniform weights
Party identification: Democrat	0.734 (0.045)	0.856 (0.045)			1.153 (0.039)
Party identification: Republican	-1.413 (0.056)	-0.717 (0.043)			-0.518 (0.030)
Party identification: other	-0.476 (0.126)	-0.468 (0.088)			-0.333 (0.056)
Party identification: not sure	-0.231 (0.080)	-0.063 (0.045)			0.019 (0.040)
Political ideology: Very liberal	0.858 (0.063)	0.780 (0.057)			0.790 (0.038)
Political ideology: liberal	0.542 (0.049)	0.511 (0.041)			0.549 (0.020)
Political ideology: conservative	-1.152 (0.058)	-1.063 (0.036)			-1.127 (0.029)
Political ideology: very conservative	-1.404 (0.098)	-1.689 (0.061)			-1.769 (0.041)
Political ideology: not sure	-0.491 (0.061)	-0.550 (0.043)			-0.533 (0.032)
Intercept (binary model)	-0.880 (0.116)				
Threshold 1 2		-2.252 (0.103)	-1.461 (0.109)	-1.775 (0.082)	-2.071 (0.076)
Threshold 2 3		-0.559 (0.099)	-0.082 (0.108)	-0.461 (0.079)	-0.310 (0.075)
Threshold 3 4		1.010 (0.101)	1.129 (0.112)	0.674 (0.082)	1.211 (0.076)
Threshold 4 5		4.110 (0.144)	3.932 (0.153)	3.370 (0.114)	4.413 (0.099)
Respondents	53661	53661	53730	53730	53661
States	50	50	50	50	50
Weighted pseudo-R-squared	0.232	0.140	0.032	0.002	0.168
Fitting sample	Complete primary sample	Complete primary sample	Demographic-complete primary sample	Outcome-valid primary sample	Complete primary sample
Weight policy	published common-content weights	published common-content weights	published common-content weights	published common-content weights	uniform weights
Recorded gender: Male (reference)	Reference	Reference	Reference		Reference
Education: high-school graduate (reference)	Reference	Reference	Reference		Reference

Term	Model 1: Binary favorable	Model 2: August labor	Model 3: Broader demographics	Model 4: Broader valid outcomes	Model 5: Uniform weights
Race and ethnicity: White (reference)	Reference	Reference	Reference		Reference
Party identification: independent (reference)	Reference	Reference			Reference
Political ideology: moderate (reference)	Reference	Reference			Reference
Probability derivatives: distinct favorable-response scale					
Average local labor derivative (SE)	0.013 (0.007)	0.019 (0.008)	-0.004 (0.010)	-0.006 (0.010)	0.016 (0.005)
Ordinary labor-derivative p	0.086	0.018	0.717	0.528	0.002
Derivative sample size	53661	53661	53730	53730	53661

Note: The binary column has an intercept rather than ordered-response thresholds. The August column changes the labor window only; wider-sample columns change completion requirements, and the uniform-weight column changes both fitting and averaging. For the shared parameter units, descriptive fit statistic and working uncertainty conventions, see Table 2. These 2012 sensitivities do not replace the primary decision rules.

Table 6. Planned sensitivity models: 2020

Term	Model 1: Binary favorable	Model 2: August labor	Model 3: Broader demographics	Model 4: Broader valid outcomes	Model 5: Uniform weights
Labor-market improvement	0.006 (0.014)		0.018 (0.013)	0.015 (0.013)	0.004 (0.012)
August labor improvement		-0.009 (0.019)			
Net-earnings contribution	-0.035 (0.017)	-0.021 (0.018)	-0.057 (0.017)	-0.045 (0.016)	-0.030 (0.014)
Property-income contribution	-0.030 (0.057)	-0.028 (0.045)	-0.020 (0.045)	0.012 (0.044)	-0.008 (0.034)
Transfer-income contribution	-0.016 (0.013)	-0.017 (0.008)	-0.033 (0.011)	-0.023 (0.010)	-0.011 (0.011)
Prior unemployment	0.042 (0.034)	0.008 (0.039)	-0.009 (0.045)	-0.032 (0.038)	0.008 (0.031)
Prior log income	-0.421 (0.256)	-0.544 (0.266)	-1.328 (0.203)	-1.304 (0.191)	-0.338 (0.180)
Age, decades centered at 45	-0.067 (0.014)	-0.098 (0.012)	-0.013 (0.013)		-0.105 (0.010)
Squared centered age	0.036 (0.005)	0.042 (0.005)	0.037 (0.005)		0.040 (0.003)
Recorded gender: female	-0.251 (0.028)	-0.169 (0.016)	-0.269 (0.016)		-0.122 (0.016)
Education: Less than high school	-0.035 (0.156)	-0.102 (0.110)	-0.120 (0.086)		-0.040 (0.068)
Education: some college	0.004 (0.049)	-0.074 (0.038)	-0.155 (0.030)		-0.075 (0.024)
Education: two-year degree	0.054 (0.060)	-0.020 (0.045)	-0.047 (0.038)		-0.026 (0.034)
Education: four-year degree	-0.105 (0.058)	-0.195 (0.036)	-0.392 (0.040)		-0.186 (0.033)
Education: postgraduate education	-0.188 (0.059)	-0.361 (0.047)	-0.752 (0.048)		-0.349 (0.033)
Race and ethnicity: Black	-0.144 (0.068)	-0.064 (0.044)	-0.697 (0.062)		0.027 (0.029)
Race and ethnicity: Hispanic	0.154 (0.074)	0.200 (0.054)	-0.182 (0.053)		0.297 (0.044)
Race and ethnicity: Asian	0.055 (0.106)	0.177 (0.067)	0.025 (0.061)		0.208 (0.060)
Race and ethnicity: Native American	-0.070 (0.168)	-0.005 (0.107)	-0.090 (0.108)		0.260 (0.084)
Race and ethnicity: mixed in 2012 or two or more races in 2020	-0.147 (0.175)	-0.121 (0.118)	-0.398 (0.128)		0.038 (0.056)
Race and ethnicity: other	0.194 (0.151)	0.271 (0.097)	0.178 (0.093)		0.316 (0.070)
Race and ethnicity: Middle Eastern	0.339 (0.334)	0.272 (0.159)	-0.059 (0.168)		0.268 (0.193)

Term	Model 1: Binary favorable	Model 2: August labor	Model 3: Broader demographics	Model 4: Broader valid outcomes	Model 5: Uniform weights
Party identification: Democrat	-1.093 (0.056)	-0.646 (0.032)			-0.529 (0.026)
Party identification: Republican	0.520 (0.054)	0.670 (0.053)			0.735 (0.037)
Party identification: other	0.045 (0.076)	-0.055 (0.061)			-0.083 (0.038)
Party identification: not sure	-0.206 (0.093)	0.035 (0.063)			0.013 (0.037)
Political ideology: Very liberal	-0.714 (0.108)	-1.059 (0.043)			-1.053 (0.033)
Political ideology: liberal	-0.660 (0.096)	-0.555 (0.044)			-0.573 (0.024)
Political ideology: conservative	0.731 (0.053)	0.713 (0.043)			0.834 (0.037)
Political ideology: very conservative	1.111 (0.045)	1.101 (0.042)			1.188 (0.035)
Political ideology: not sure	-0.196 (0.099)	-0.153 (0.060)			-0.138 (0.036)
Intercept (binary model)	-1.209 (0.164)				
Threshold 1 2		-0.781 (0.147)	-1.312 (0.133)	-0.857 (0.116)	-0.536 (0.114)
Threshold 2 3		0.741 (0.148)	-0.064 (0.133)	0.351 (0.116)	0.990 (0.115)
Threshold 3 4		1.377 (0.148)	0.476 (0.134)	0.876 (0.116)	1.625 (0.117)
Threshold 4 5		2.453 (0.156)	1.430 (0.144)	1.814 (0.127)	2.707 (0.126)
Respondents	58272	58272	58305	58305	58272
States	50	50	50	50	50
Weighted pseudo-R-squared	0.172	0.105	0.018	0.003	0.110
Fitting sample	Complete primary sample	Complete primary sample	Demographic-complete primary sample	Outcome-valid primary sample	Complete primary sample
Weight policy	published common-content weights	published common-content weights	published common-content weights	published common-content weights	uniform weights
Recorded gender: Male (reference)	Reference	Reference	Reference		Reference
Education: high-school graduate (reference)	Reference	Reference	Reference		Reference

Term	Model 1: Binary favorable	Model 2: August labor	Model 3: Broader demographics	Model 4: Broader valid outcomes	Model 5: Uniform weights
Race and ethnicity: White (reference)	Reference	Reference	Reference		Reference
Party identification: independent (reference)	Reference	Reference			Reference
Political ideology: moderate (reference)	Reference	Reference			Reference
Probability derivatives: distinct favorable-response scale					
Average local labor derivative (SE)	0.001 (0.002)	-0.001 (0.003)	0.003 (0.002)	0.003 (0.002)	0.000 (0.002)
Ordinary labor-derivative p	0.672	0.619	0.180	0.269	0.754
Derivative sample size	58272	58272	58305	58305	58272

09

Note: These 2020 specifications address threshold, labor timing, response completion and weighting choices separately. The sample sizes identify each fitting population; the August specification retains second-quarter income, and uniform weights apply to estimation and probability averaging. Coefficients remain distinct from the final local-derivative block, whose ordinary probabilities are not adjusted family decisions. Statistical definitions common to the model families appear beneath Table 2.

Table 7. Primary comparisons and their inferential qualifications

A. Directional comparisons: economic evaluations

Election	Variable	Derivative (SE)	98.75% interval	Adjusted p
2012	Labor improvement	0.014 (0.006)	[-0.001, 0.030]	0.075
2012	Net earnings	0.003 (0.002)	[-0.002, 0.009]	0.522
2020	Labor improvement	0.002 (0.002)	[-0.003, 0.006]	1.000
2020	Net earnings	-0.005 (0.002)	[-0.012, 0.001]	0.183

B. Joint exact common-coefficient restrictions

Election	F statistic	Numerator df	Denominator df	Ordinary p	Holm p
2012	0.354	2	49	0.703	0.960
2020	0.745	2	49	0.480	0.960

C. Component contrasts and state-deletion sensitivity

Election	Contrast	Derivative difference	Primary interval	Primary classification	Jackknife interval	Jackknife classification
2012	Net earnings – Property income	0.005	[-0.014, 0.025]	Unresolved	[-0.018, 0.029]	Unresolved
2012	Net earnings – Transfers	0.005	[-0.029, 0.038]	Unresolved	[-0.037, 0.046]	Unresolved
2012	Property income – Transfers	-0.001	[-0.038, 0.037]	Unresolved	[-0.047, 0.046]	Unresolved
2020	Net earnings – Property income	-0.003	[-0.023, 0.016]	Unresolved	[-0.029, 0.022]	Unresolved
2020	Net earnings – Transfers	-0.003	[-0.011, 0.004]	Practically similar	[-0.013, 0.006]	Practically similar

Election	Contrast	Derivative difference	Primary interval	Primary classification	Jackknife interval	Jackknife classification
2020	Property income – Transfers	-0.000	[-0.017, 0.017]	Practically similar	[-0.024, 0.024]	Unresolved

D. Secondary presidential-approval derivatives

Election	Variable	Derivative (SE)	Holm p
2012	Labor improvement	0.002 (0.006)	1.000
2012	Net earnings	0.002 (0.002)	1.000
2012	Property income	-0.007 (0.006)	1.000
2012	Transfers	-0.014 (0.012)	1.000
2020	Labor improvement	0.001 (0.001)	1.000
2020	Net earnings	-0.006 (0.002)	0.059
2020	Property income	-0.011 (0.005)	0.295
2020	Transfers	-0.002 (0.002)	0.986

Note: All quantities are conditional associations. Panel A uses the four-comparison Bonferroni family. Panel B uses Holm adjustment across two joint restrictions; nonrejection is not evidence of similarity. Panel C uses 99.167% intervals (the exact nominal level is one minus .05 divided by six) and a fixed ± 0.020 probability-derivative margin; classifications use unrounded intervals and strict containment. Jackknife intervals retain the original fifty-state averaging distribution while refitting after each state deletion.

Note: The four positive primary associations are not jointly established. Neither election establishes practical similarity across all three components, and the combined six-comparison similarity condition is not established. Panel D uses a separate eight-comparison Holm family and separate approval samples; none of these secondary associations is established at the .05 family threshold. The complete export retains all 24 decision records and all sensitivity contrasts.

Table 8. Response accounting, exclusions and retention

A. Analytic sample descent

	Election	Stage	Respondents	Eligible weight retained	States	Weight effective sample size
2012	2012	Eligible sample	54,426	100.00%	50	22888.96
	2012	Valid economic evaluation	53,730	98.73%	50	22615.53
	2012	Demographic-complete economic sample	53,730	98.73%	50	22615.53
	2012	Complete economic sample	53,661	98.57%	50	22604.91
	2012	Valid presidential approval	53,214	97.87%	50	22415.77
	2012	Demographic-complete approval sample	53,214	97.87%	50	22415.77
	2012	Complete approval sample	53,152	97.72%	50	22407.84
	2020	Eligible sample	60,803	100.00%	50	28086.87
	2020	Valid economic evaluation	58,305	94.51%	50	27588.64
	2020	Demographic-complete economic sample	58,305	94.51%	50	27588.64
2020	2020	Complete economic sample	58,272	94.46%	50	27567.69
	2020	Valid presidential approval	59,311	96.71%	50	27831.95

Election	Stage	Respondents	Eligible weight retained	States	Weight effective sample size
2020	Demographic-complete approval sample	59,311	96.71%	50	27831.95
2020	Complete approval sample	59,277	96.63%	50	27833.05

B. Outcome response accounting among eligible respondents

Election	Outcome	Response category	Count	Weighted share
2012	Economic evaluation	Substantive response	53,730	98.73%
2012	Economic evaluation	Uncertain response	664	1.23%
2012	Economic evaluation	Unanswered or not asked	32	0.04%
2012	Presidential approval	Substantive response	53,214	97.87%
2012	Presidential approval	Uncertain response	1,157	2.02%
2012	Presidential approval	Unanswered or not asked	55	0.11%
2020	Economic evaluation	Substantive response	58,305	94.51%
2020	Economic evaluation	Uncertain response	2,389	5.28%
2020	Economic evaluation	Unanswered or not asked	109	0.21%
2020	Presidential approval	Substantive response	59,311	96.71%
2020	Presidential approval	Uncertain response	1,460	3.24%
2020	Presidential approval	Unanswered or not asked	32	0.05%

C. Eligibility exclusions

Nonexclusive exclusion reason	2012	2020
Unresolved contrary age evidence	0	0
Invalid or nonpositive weight	0	0
Unresolved missing interview date	0	0
Unmatched context	109	197
Missing state	0	0
District of columbia	109	197
Other nonstate residence	0	0
Unresolved state code	0	0

D. Conditional completion and unanswered-item bounds

Election	Outcome	Weight retention among substantive respondents	Lower favorable prevalence	Upper favorable prevalence
2012	Economic evaluation	99.84%	32.14%	32.18%
2012	Presidential approval	99.84%	52.19%	52.29%
2020	Economic evaluation	99.95%	21.55%	21.76%
2020	Presidential approval	99.92%	44.48%	44.54%

Note: Eligibility-exclusion flags overlap and must not be added. Invalid weights do not contribute to weight masses. Conditional completion percentages use valid substantive responses as their denominator; they differ from retention of the full eligible weight shown in Panel A.

Note: Uncertain responses remain observed responses, not neutral categories or genuinely unanswered items. Favorable-prevalence bounds vary only genuinely unanswered eligible responses while retaining observed uncertainty in the denominator; they are neither slope bounds nor forced-choice bounds. The local supplement retains native response categories and covariate-specific missingness.

Table 9. Support and state-deletion diagnostics

A. Smallest state samples

Election	Outcome sample	States	Minimum respondents	Minimum weight effective sample size
2012	Economic evaluation	50	104	53.42
2012	Presidential approval	50	101	51.92
2020	Economic evaluation	50	93	48.63
2020	Presidential approval	50	91	50.31

B. Economic design support

Election	Weighting	Rank	Maximum VIF	Condition number	Maximum leverage	Effective states
2012	Equal state	7	1.60	2.31	0.68	50.00
2012	Survey state weight	7	1.76	2.39	0.57	26.95
2020	Equal state	7	2.65	3.19	0.49	50.00
2020	Survey state weight	7	2.43	3.27	0.51	25.15

C. Full-design conditional economic variation

Election	Variable	Respondent residual SD	State-mean residual SD	Largest state residual share	One unit / residual SD
2012	Labor improvement	0.48	0.48	23.20%	2.11
2012	Net earnings	1.06	1.78	15.06%	0.94
2012	Property income	0.49	0.75	14.80%	2.05
2012	Transfers	0.22	0.25	12.34%	4.51

Election	Variable	Respondent residual SD	State-mean residual SD	Largest state residual share	One unit / residual SD
2020	Labor improvement	1.67	2.11	17.59%	0.60
2020	Net earnings	0.99	1.22	19.65%	1.01
2020	Property income	0.40	0.48	27.15%	2.51
2020	Transfers	1.45	1.76	15.32%	0.69

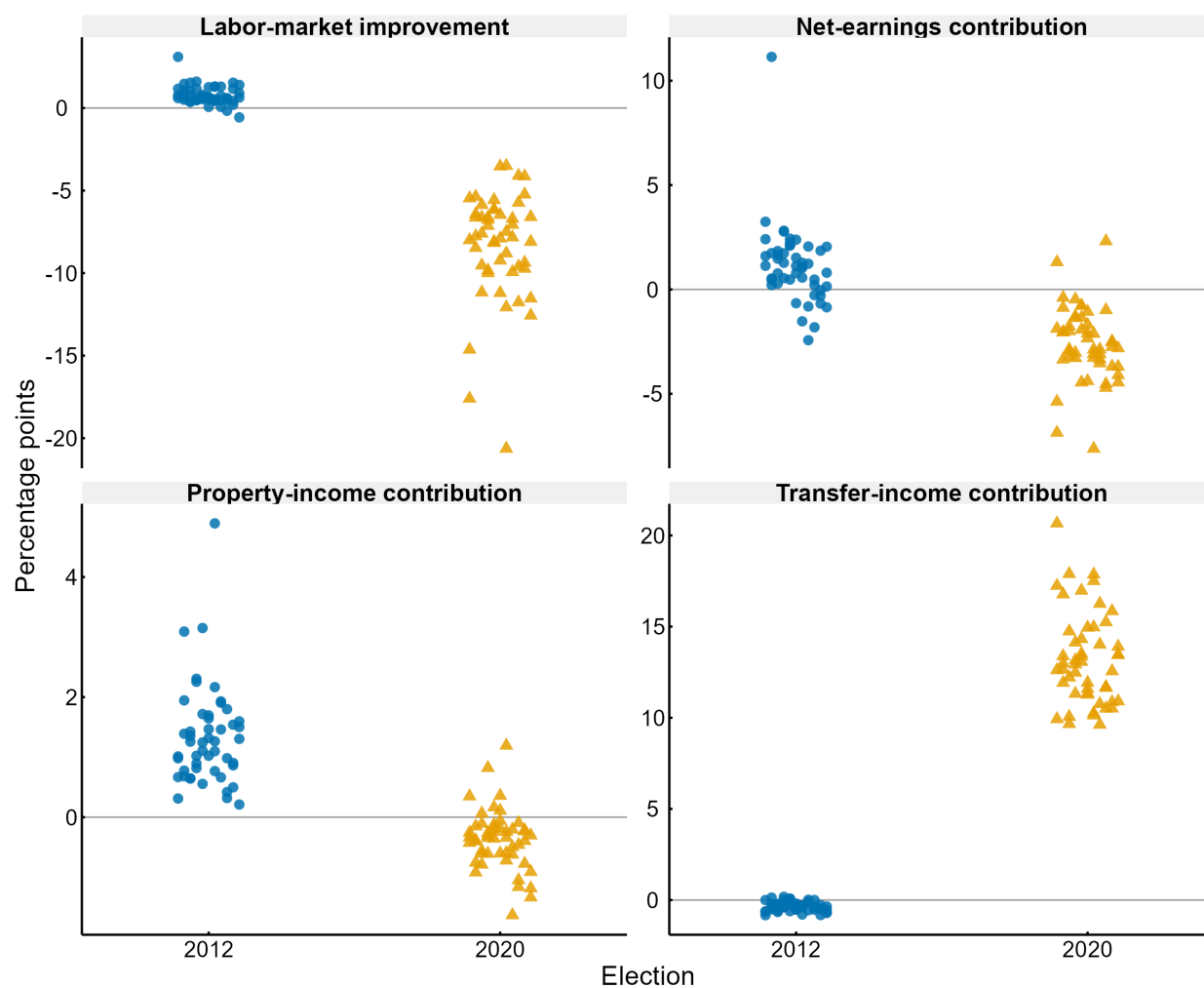
D. Leave-one-state-out fits

Election	Attempted	Succeeded	Fitting states	Averaging states	Fits with warnings
2012	50	50	49	50	0
2020	50	50	49	50	0

Note: VIFs and condition numbers describe the economic design under the indicated weighting. Full-design residuals also condition on respondent covariates; state-mean residual dispersion is not the respondent-level residual dispersion. Residual-share and leverage diagnostics are descriptive, not tests of an identified mechanism.

Note: Each deletion fit uses 49 states but evaluates the derivative over the original 50-state distribution. All attempts, deletion-specific estimates, score-norm concentration and complete fit-statistic records remain available in the local supplement. Score-norm concentration depends on coefficient and threshold units and is not an invariant measure of inferential information. Weight effective sample sizes do not replace the number of state contexts.

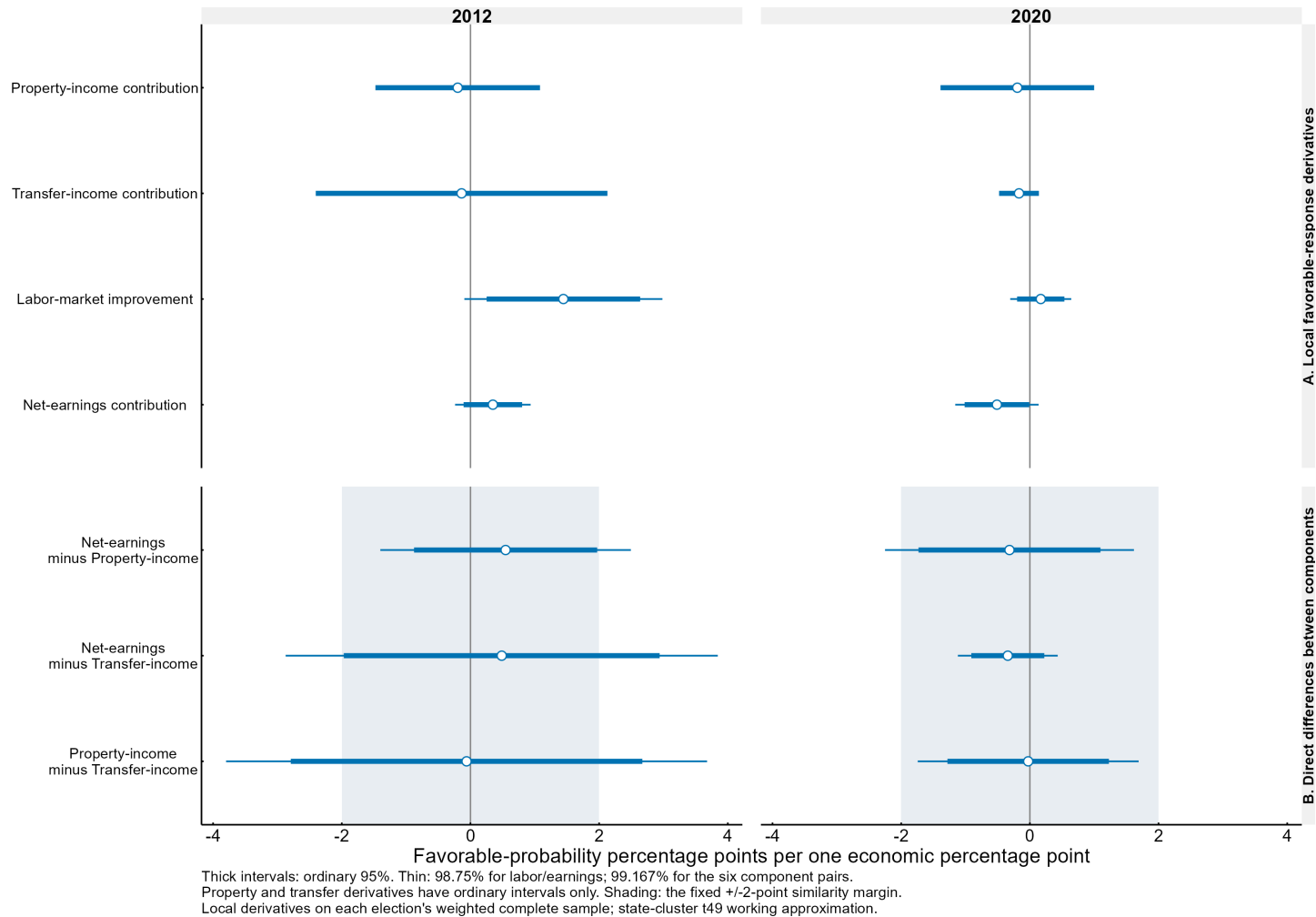
Figure 1. State labor conditions and common-scale income contributions



Each point is one state (50 per election). Labor: prior minus current April-June unemployment.
Income: annual Q2 real per-capita change divided by prior total income; revised source vintage.

Note: Points summarize the 50 states in each election context. Labor improvement is the prior-year minus election-year second-quarter unemployment rate. Earnings, property income and transfers are percentage-point contributions to real per-capita total-income growth from the prior second quarter. The common denominator preserves component additivity; the units do not represent equal dollar changes across states.

Figure 2. Local probability derivatives and direct income-component comparisons



Note: The figure reports local derivatives for favorable national economic evaluations and direct pairwise differences in the primary component models. Ordinary intervals are shown alongside the distinct adjusted interval families. The reference bands for component differences mark the study's fixed practical-similarity margin. Quantities are local associations on the observed election-specific grid, not finite policy changes. The state-deletion qualification to the 2020 property-transfer comparison appears in Table 7.

Appendix: Measurement and Supplementary Evidence

The annual questionnaires ask respondents whether, over the past year, the nation’s economy has improved or deteriorated (Ansolabehere and Schaffner 2013; Schaffner, Ansolabehere, and Luks 2021). The ordered substantive categories distinguish much worse, worse, about the same, better and much better; the 2020 intermediate response labels use the qualifier “somewhat,” whereas the 2012 labels do not. The analysis preserves the ordered positions and keeps uncertainty outside that substantive ordering. Approval distinguishes strong and weaker approval or disapproval of the president’s job performance. Table 1 shows the corresponding distributions within each separately constructed sample, including availability of the other outcome. This harmonization aligns codes and ordering, without assuming that respondents in different electoral circumstances attach identical meanings to each position.

The economic accounts use BEA’s quarterly state personal-income table SQINC4: line 10, total income; line 20, population; line 45, net earnings by residence; line 46, dividends, interest and rent; and line 47, current transfers. The archive was acquired from the official BEA quarterly state-income endpoint on October 1, 2026, with an HTTP modification date of September 30, 2026. Prices use BEA’s all-items personal-consumption-expenditures series DPCERG, National Income and Product Accounts (NIPA) Table 2.3.4, quarterly sheet T20304-Q, with 2017=100. The official BEA Section 2 workbook was acquired on October 1, 2026; its HTTP modification date and the price-index sheet’s publication date are September 30, 2026. These acquired files supply the September 2026 release vintage; the cited 2020 BEA release supplies accounting definitions and the earnings qualification, not the analysis vintage.

The income comparisons use the second quarters of 2011 and 2012, and of 2019 and 2020. Annual-

rate income is divided by contemporaneous population and deflated consistently across components; it is not divided by four. Labor measures use BLS Local Area Unemployment Statistics (LAUS) seasonally adjusted statewide monthly unemployment rates, measure 03. The fifty-state series use the LASST prefix, two-digit state FIPS code, eleven zeroes and suffix 03; the requested span was 2011–2020. The responses were retrieved through the official BLS public time-series API on October 1, 2026. April–June averages define the primary labor comparison, and August rates define its single timing sensitivity. The same acquired vintage is retained within each source. These measurements describe reconstructed contexts rather than real-time information sets.

Tables 3–6 provide the full adjustment, approval and sensitivity specifications. Table 7 separates the directional family, common-coefficient restrictions, practical component comparisons and approval family. Table 8 preserves outcome uncertainty and nonexclusive exclusion accounting instead of treating every omitted response as interchangeable. Table 9 summarizes state support and the complete state-deletion exercise. The accompanying local supplementary tables retain full coefficient probabilities, state-specific diagnostics, deletion estimates, excluded-case distributions and source documentation beyond these summaries. Their additional detail permits inspection of the information behind the comparisons; it does not change the inferential populations, multiply the number of independent economic contexts or constitute additional hypothesis tests.